Response to the Associate Editor and Reviewers

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Title: Measurement-Aware Ellipse Detection for Real-Time Wear Assessment of Roller Tank Lugs

Response to the Associate Editor

Response We sincerely thank the Associate Editor for coordinating the review of our manuscript and giving us the opportunity to revise it. We also thank all reviewers for their detailed and constructive comments. We carefully examined every comment and revised the manuscript accordingly. The main changes are summarized in three areas.

(1) We clarified the method and its contributions. We do not claim that the ellipse representation or GWD itself is new. The contribution lies in combining direct ellipse prediction, camera-based physical-size reconstruction, measurement-aware supervision, GCER boundary refinement, and multi-frame fusion into a complete method for millimeter-level size measurement and wear assessment.

(2) We expanded the data description and experimental validation. We now explain how the dataset was partitioned, how the ellipse labels were generated and manually checked, and how the physical reference dimensions were measured. We also added comparisons with recent detectors and segmentation-based measurement, together with experiments on loss-weight sensitivity, camera-to-target distance, distance perturbation, side-projection correction, GCER boundary quality, samples near the maintenance thresholds, Kalman filtering, computational complexity, and latency on the mine-site server.

(3) We reviewed the language and technical presentation throughout the manuscript. We corrected the English, standardized terminology, abbreviations, symbols, and equation descriptions, and stated more clearly the current limitations under severe occlusion, extremely low illumination, and cross-mine application. The corresponding changes and supporting evidence are described in the point-by-point responses below.

Response to Reviewer 1

R1.General The paper studies roller tank lug wear assessment using an ellipse aware YOLOv5 detector, camera metrology, edge guided contour refinement, and temporal size fusion. I recommend major revision.

Response We sincerely thank the reviewer for the careful evaluation and for recommending a major revision. We agree that the original manuscript required substantially stronger methodological clarification and experimental support. In response, we clarified the measurement-oriented role of the ellipse prediction head and revised the loss and metrology descriptions to define their variables, assumptions, and scope. We documented the dataset partitioning and physical ground-truth procedure and added targeted evaluations across distance, distance uncertainty, off-axis displacement, GCER boundary quality, threshold neighborhoods, and multi-frame fusion.

R1.Q1 The ellipse aware head is a useful representation, but its novelty over rotated detection and segmentation based measurement is not separated cleanly.

Response We thank the reviewer for identifying this important positioning issue. We agree that the original manuscript did not separate the role of the ellipse-aware head sufficiently clearly from established rotated detection and segmentation-based measurement. In particular, the previous wording could be read as claiming novelty for ellipse representation itself. We have revised the manuscript to remove this ambiguity. Neither the ellipse representation nor GWD is claimed as a new geometric primitive or loss formulation in this work.

Rotated detectors generally predict an oriented enclosing rectangle and optimize category recognition and localization overlap. Such a representation is effective for oriented target localization, but the dimensions of the enclosing box are not explicitly optimized according to their consequences for physical size reconstruction. Segmentation-based measurement methods provide dense object masks and then extract, fit, and calibrate their boundaries. They can represent irregular contours, but the resulting physical measurement depends on mask completeness and boundary quality, which are difficult to maintain under dust, clutter, and partial occlusion.

The role of our ellipse-aware head is therefore more specific than merely replacing a rectangular box with another detection representation. It predicts a compact set of contour parameters whose semi-axes are passed directly to a differentiable analytical camera-metrology mapping. The measurement-aware loss then supervises these parameters according to the millimeter-level errors that they produce after physical reconstruction. GWD is used as an established joint geometric regularizer, while GCER subsequently searches for local boundary evidence around the predicted ellipse and robustly refits it without introducing a second dense segmentation network. The methodological contribution is this coupling of learned ellipse localization, physical-size supervision, and lightweight geometric refinement.

We have clarified this distinction in the Introduction, the task formulation, the first two contribution

statements, and the dimension-measurement analysis. The component-wise results also separate the sources of improvement. Changing from YOLOv5 box-derived size estimation to direct ellipse parameterization reduced the overall MAE from 16.36 to 6.18 mm. GWD, measurement-aware supervision, and GCER then reduced it successively to 5.34, 4.33, and 3.17 mm. Thus, the final improvement is not attributed solely to the ellipse representation.

YOLO and YOLOv7 established fast single-stage object detection, and subsequent YOLO-based models have been applied to industrial defect inspection; DETR offers an alternative end-to-end detection formulation. However, these detectors generally output rectangular boxes intended to locate an object rather than recover the physical contour of a roller tank lug. Because the lug appears circular or elliptical in the image, a rectangular box necessarily includes additional background. Under partial occlusion or off-axis imaging, the box may also change with the visible region. A lug can therefore be detected correctly while its box-derived dimensions remain inaccurate, which may subsequently alter the assigned wear state. Roller tank lug monitoring thus requires a target representation that directly describes the working contour and supports physical size reconstruction.

An ellipse is not itself a new geometric representation. The modification introduced here is that the detection head directly outputs parameters compatible with size measurement. In particular, a and b describe the semi-axes of the lug contour rather than the side lengths of an enclosing oriented rectangle, and the decoded semi-axes are passed directly to the camera-metrology mapping for physical-size reconstruction.

(1) We propose an ellipse-aware prediction head that directly outputs the lug center, semi-major axis, semi-minor axis, and orientation, allowing the detection result to be used directly for physical size reconstruction.

(2) We combine GWD, camera-based size reconstruction, and a measurement-aware loss so that ellipse training accounts for both image-domain localization error and its effect on millimeter-level dimensions.

The largest reduction therefore came from replacing box-derived dimensions with direct ellipse prediction. GWD, $\mathcal{L}_{\mathrm{ma}}$, and GCER subsequently addressed parameter coupling, millimeter-level measurement error, and local boundary offset. When little contour remains visible, GCER has less boundary information and provides a smaller correction.

R1.Q2 Related work for geometric visual measurement may include "Grape Berry Detection and Size Measurement Based on Edge Image Processing and Geometric Morphology."

Response We thank the reviewer for directing us to this relevant geometric visual-measurement study. We agree that the original manuscript focused too narrowly on mining inspection and did not sufficiently discuss its connection with geometric visual measurement in other application domains. We have carefully read and cited the suggested work by Luo et al. In that study, Canny edge extraction, concave-point detection, and contour-segment reorganization were combined to recover

individual grape berry contours and estimate their physical size. The study demonstrates that edge information and contour geometry can provide a clear and interpretable basis for physical-size measurement, and it offers valuable reference for the present work.

We have incorporated this study into the Introduction together with a recent learning-based crack-width measurement method. Considering the coal-dust interference, background clutter, side-surface interference, and partially missing contours present in mine images, the proposed framework uses a learned global ellipse to determine the overall target shape, followed by local boundary refinement and camera-based physical-size reconstruction. The revised Introduction now places the study within the broader context of geometric visual measurement and explains the application conditions and technical route addressed in this work.

In machine-vision-based dimensional measurement, a common method is to extract the target boundary from the image and then calculate its physical dimensions using imaging geometry. Luo et al. measured grape berry size using edge detection and contour reorganization, while Wu et al. combined crack segmentation, maximum-inscribed-circle estimation, and depth-camera geometry to calculate crack width. These studies demonstrate that edge and contour geometry provide a clear and interpretable basis for physical-size measurement. Under coal dust, background clutter, or partial occlusion, however, target boundaries may become fragmented or displaced, which subsequently affects contour fitting and size reconstruction. Considering these characteristics of field images of roller tank lugs, the present framework uses a learned global ellipse to represent the overall target shape and then combines local boundary refinement with a camera model for physical-size measurement.

R1.Q3 The physical ground truth for lug width and height is weakly documented, leaving calibration error and manual measurement variance outside the reported MAE.

Response We thank the reviewer for identifying the insufficient description of the physical ground-truth acquisition and its measurement variability. We agree that the original manuscript did not clearly distinguish the uncertainty of the reference measurements from the end-to-end measurement error of the proposed system.

The physical reference dimensions were obtained independently of the image annotations. The major-axis dimension W_{gt} and minor-axis dimension H_{gt} of each physical roller tank lug were measured by three field staff members, with each operator performing ten repeated measurements. For the lug dimensions, a rigid plate was placed against one side of the roller tank lug as a reference surface, and laser range measurements were taken from the opposite side. A laser distance meter with a manufacturer-specified accuracy of ± 1.5 mm was used as the primary instrument, while the resulting dimensions were cross-checked with a large-range industrial vernier caliper. The camera-to-target distance d was measured by the laser distance meter at the corresponding acquisition position. The averages of the repeated measurements were used as W_{gt}, H_{gt}, and d.

To quantify measurement repeatability, we additionally evaluated a representative roller tank lug

using 30 measurements obtained by three operators. The measured major axis, minor axis, and camera-to-target distance were 372.73 ± 0.51 mm, 369.41 ± 0.49 mm, and 2349.94 ± 1.32 mm, respectively. The corresponding measurement ranges were 1.93, 1.58, and 4.49 mm.

We further clarify that the reported MAE was calculated by directly comparing the reconstructed physical dimensions with the averaged repeated measurements. It therefore quantifies the discrepancy of the complete measurement pipeline relative to this experimentally established physical reference. Errors arising from ellipse localization, camera calibration, camera-to-target distance measurement, and physical-size reconstruction can all contribute to this discrepancy. We therefore interpret the MAE as an end-to-end error relative to the averaged physical reference, rather than as an isolated neural-network regression error or a complete uncertainty-budget decomposition. We have revised Section IV-B (Data Source and Dataset Composition) to document the physical ground-truth acquisition procedure and the error definition.

The reference dimensions W_{gt} and H_{gt} and the distance d were measured by three field operators, each taking ten readings. For the lug dimensions, a rigid plate was placed against one side of the roller tank lug as a reference surface, and laser range measurements were taken from the opposite side. A laser distance meter with a specified accuracy of ± 1.5 mm was the primary instrument, and the resulting dimensions were cross-checked with a large-range industrial vernier caliper. The averages were used as ground truth, and the reported MAE was calculated directly against them.

R1.Q4 The image split appears image based, and frames from the same lug or video could enter different subsets (p. 7).

Response We thank the reviewer for raising the possibility of information leakage between dataset subsets. We agree that the original manuscript reported only the image counts and split ratio, without defining the grouping unit used for partitioning. The dataset was not divided by randomly assigning individual images or video frames. Before subset allocation, all samples were grouped by physical roller tank lug identity, acquisition period, and source video or continuous acquisition sequence. Each group was assigned in its entirety to exactly one subset. Consequently, no physical lug, source video, duplicate image, temporally adjacent frame, or near-duplicate sequence was shared among the training, validation, and test sets. This group-wise procedure excludes object-identity and temporal-frame leakage by construction. We have added the procedure to Section IV-B (Data Source and Dataset Composition).

To prevent adjacent frames or the same physical lug from appearing in different subsets, samples were grouped by lug identity, acquisition period, and source video, and each group was assigned to only one subset. The 720 test images came from 720 different videos; the complete videos were retained separately for temporal evaluation.

R1.Q5 Camera distance is treated as recorded input, but the distance sensor, calibration procedure, and sensitivity to distance noise are not quantified.

Response We thank the reviewer for identifying the insufficient documentation of camera calibration, distance acquisition, and distance-noise sensitivity. The camera-to-target distance d was measured at the corresponding acquisition position using a laser distance meter with a manufacturer-specified accuracy of ± 1.5 mm. As clarified in our response to R1.Q3, repeated field measurements were averaged and associated with the corresponding acquisition group, and the lug dimensions were cross-checked with a large-range industrial vernier caliper. Multiple fixed-focus cameras were installed at different positions and heights to monitor roller tank lugs on different hoisting conveyances. Each camera, lens, and resolution configuration was calibrated separately using Zhang's method before acquisition. Neither optical nor digital zoom was used, and each sample was reconstructed using the intrinsics of its corresponding source camera.

We additionally performed a distance-perturbation experiment on the test set. For each test sample, the recorded distance was perturbed as $d' = d + \varepsilon$, where $\varepsilon \sim \mathcal{U}(-\delta, \delta)$ and $\delta \in \{1.5, 3, 5\}$ mm. The detector outputs, camera intrinsic parameters, and physical ground truth were held fixed so that only sensitivity to d was evaluated. Each perturbation level was evaluated using 30 independent perturbation realizations. At the rated uncertainty of ± 1.5 mm, the overall MAE, RMSE, and MAPE changed from 3.170 mm, 3.180 mm, and 0.840% to 3.192 ± 0.015 mm, 3.205 ± 0.021 mm, and $0.855 \pm 0.008\%$, respectively. Under the stronger ± 3 mm disturbance, the corresponding results were 3.242 ± 0.031 mm, 3.261 ± 0.033 mm, and $0.876 \pm 0.012\%$. Even under the ± 5 mm stress test, which exceeds three times the rated uncertainty of the distance meter, the results remained 3.292 ± 0.047 mm, 3.310 ± 0.047 mm, and $0.900 \pm 0.024\%$.

Relative to the unperturbed result, the mean MAE increased by only 0.022 mm (0.69%) at the rated uncertainty of ± 1.5 mm. The corresponding increases were 0.072 mm (2.27%) at ± 3 mm and 0.122 mm (3.85%) at ± 5 mm. RMSE and MAPE followed the same monotonic trend, while their standard deviations also increased gradually with perturbation magnitude. This behavior is consistent with the first-order propagation relationship $\Delta W/W \approx \Delta H/H \approx \Delta d/d$ derived from the pinhole reconstruction equations. Thus, uncertainty in d is measurable, but the rated sensor uncertainty contributes only a small additional error relative to the end-to-end baseline error. Even the ± 5 mm stress test, which is more than three times the rated uncertainty, produced progressive rather than abrupt degradation. These results directly quantify the operating tolerance of the reconstruction stage rather than merely asserting robustness. We revised Section IV-A to document the on-site calibration and fixed-focus acquisition protocol, while Section IV-D now reports the perturbation setup and key result concisely; the complete quantitative results are retained here.

Multiple fixed-focus cameras were installed at different positions and heights to monitor roller tank lugs on different hoisting conveyances. Each camera, lens, and resolution configuration was calibrated separately using Zhang's method. The focal length and image resolution of each camera remained fixed during its acquisition sessions, and each sample was reconstructed using the intrinsics of its corresponding source camera.

To examine distance uncertainty, we added uniform perturbations of ± 1.5, ± 3, and ± 5 mm to

the recorded distance while fixing the detections, camera intrinsics, and physical references; each level was repeated 30 times. MAE increased from 3.170 mm without perturbation to 3.192 mm at the rated ±1.5 mm uncertainty and 3.292 mm at ±5 mm. The gradual increase agrees with $\Delta W/W \approx \Delta H/H \approx \Delta d/d$.

Table VI. Sensitivity of physical-size reconstruction to camera-to-target distance perturbations.

Distance perturbation	Overall MAE / mm	RMSE / mm	MAPE / %
Recorded distance (0 mm)	3.170	3.180	0.840
Rated uncertainty (±1.5 mm)	3.192 ± 0.015	3.205 ± 0.021	0.855 ± 0.008
Moderate noise (±3 mm)	3.242 ± 0.031	3.261 ± 0.033	0.876 ± 0.012
Stress test (±5 mm)	3.292 ± 0.047	3.310 ± 0.047	0.900 ± 0.024

R1.Q6 The side projection correction uses offsets and lug thickness, while the acquisition protocol gives little evidence that these variables are measured per frame.

Response We thank the reviewer for identifying that the sources of the offset and thickness variables were not sufficiently documented. The offsets m_x and m_y are not manually measured for each frame. For a refined ellipse center (u, v), they are calculated frame by frame from the recorded camera-to-target distance d and the calibrated intrinsic parameters (f_x, f_y, c_x, c_y) as $m_x = d(u - c_x)/f_x$ and $m_y = d(v - c_y)/f_y$. The lug thickness t is a manufacturer-specified structural dimension. Because tread wear changes the outer diameter rather than this structural thickness, t was treated as a fixed constant for the lug type used in the experiment. We have added these parameter sources and assumptions to Section III-A.

We also added a paired correction-on/correction-off experiment using 50 samples of the 360-mm lug specification. The detector output, camera intrinsics, recorded distance, and physical ground truth were identical within each pair, so the comparison isolated only the side-projection correction. The effective off-axis angle was defined as $\beta = \arctan(\sqrt{m_x^2 + m_y^2}/d)$. The observed range was 0.307°–5.335°. Enabling the correction reduced the overall MAE from 5.169 to 3.347 mm, RMSE from 5.770 to 3.371 mm, and MAPE from 1.436% to 0.930%. When the samples were divided into four approximately equal-sized groups after ordering by β, the average error decreased in every group, and the MAE reduction increased from 10.63% in the lowest-angle group to 48.30% in the highest-angle group. This trend directly supports the role of the correction in compensating for center displacement from the optical axis.

We have also clarified the scope of this model. The field installation enforces near-frontal acquisition, and β describes the angle between the target-center viewing ray and the optical axis. It is not the in-plane ellipse orientation θ and does not represent an arbitrary three-dimensional tilt of the lug plane. The revised manuscript now states the acquisition scope and the key paired result concisely, while the complete protocol and grouped results are provided here.

The field cameras were installed at different monitoring positions for near-frontal imaging, with the

lug plane approximately parallel to the image plane of the corresponding camera. The correction therefore addresses the side-projection error caused by displacement of the target center from the optical axis. Let (u, v) denote the refined ellipse center and (c_x, c_y) the principal point obtained from the calibration of the corresponding camera. The horizontal and vertical physical offsets are calculated for each frame as

$$m_x = \frac{d(u - c_x)}{f_x}, \qquad m_y = \frac{d(v - c_y)}{f_y}.$$

Thus, m_x and m_y are derived from the detected center, calibrated intrinsics, and recorded distance; they are not measured manually for each frame. The lug thickness t is a manufacturer-specified structural constant and does not change with side-surface wear.

Paired evaluation of side-projection correction. Error entries are reported as correction OFF/ON.

Group	Observed β / deg	n	MAE / mm	RMSE / mm	MAPE / %	MAE reduction / %
Q1	0.307–1.280	13	3.537/3.161	3.738/3.171	0.982/0.878	10.63
Q2	1.406–2.563	12	4.978/3.356	5.282/3.373	1.383/0.932	32.57
Q3	2.579–3.352	12	5.451/3.405	6.039/3.432	1.514/0.946	37.54
Q4	3.561–5.335	13	6.718/3.473	7.394/3.504	1.866/0.965	48.30
Overall	0.307–5.335	50	5.169/3.347	5.770/3.371	1.436/0.930	35.24

A paired correction-on/off test used 50 near-frontal samples with target-center viewing angles of 0.307°–5.335°. With identical detections, camera parameters, distances, and physical references, the side-projection correction reduced overall MAE from 5.169 to 3.347 mm.

R1.Q7 Loss weights and the sensitivity weighting parameter are fixed empirically, with no sweep showing whether the measurement aware term is stable across distances.

Response We thank the reviewer for identifying the need to justify the loss coefficients and to evaluate the measurement-aware term across camera-to-target distances. We agree that reporting fixed empirical coefficients without a quantitative sensitivity analysis was insufficient. The standard YOLOv5 training hyperparameters were searched using its hyperparameter-evolution strategy. However, λ_{gwd}, λ_{ma}, and α are newly introduced measurement-related coefficients and are not part of the standard YOLOv5 search space. We therefore evaluated them separately by one-factor-at-a-time sensitivity analysis. When λ_{gwd} was scanned, $\lambda_{\text{ma}} = 0.002$ and $\alpha = 0.001$ were fixed. When λ_{ma} was scanned, $\lambda_{\text{gwd}} = 0.05$ and $\alpha = 0.001$ were fixed. When α was scanned, $\lambda_{\text{gwd}} = 0.05$ and $\lambda_{\text{ma}} = 0.002$ were fixed. The dataset split, input resolution, training schedule, detector architecture, and GCER configuration were unchanged. Each setting was independently repeated three times, and the mean and standard deviation of MAE were reported.

For $\lambda_{\text{gwd}} \in \{0.010, 0.025, 0.050, 0.100\}$, the corresponding overall MAEs were 3.29 ± 0.052, 3.22 ± 0.044, 3.17 ± 0.039, and 3.25 ± 0.047 mm, respectively. For $\lambda_{\text{ma}} \in \{0.0005, 0.0010, 0.0020, 0.0050\}$,

the overall MAEs were 3.31 ± 0.055, 3.23 ± 0.046, 3.17 ± 0.039, and 3.28 ± 0.053 mm. For $\alpha \in \{0, 0.0005, 0.0010, 0.0020, 0.0050\}$, the overall MAEs were 3.27 ± 0.051, 3.21 ± 0.044, 3.17 ± 0.039, 3.20 ± 0.043, and 3.30 ± 0.057 mm. The final configuration, $\lambda_{\text{gwd}} = 0.05$, $\lambda_{\text{ma}} = 0.002$, and $\alpha = 0.001$, produced the lowest mean MAE among the tested coordinate-wise settings. Because the one-factor-at-a-time analysis did not enumerate all joint combinations, we do not describe it as a global optimum.

Sensitivity analysis of the GWD, measurement-aware, and sensitivity-term coefficients.

Scanned coefficient	Value	Overall MAE / mm
λ_{gwd}	0.010	3.29 ± 0.052
λ_{gwd}	0.025	3.22 ± 0.044
λ_{gwd}	**0.050**	$\mathbf{3.17 \pm 0.039}$
λ_{gwd}	0.100	3.25 ± 0.047
λ_{ma}	0.0005	3.31 ± 0.055
λ_{ma}	0.0010	3.23 ± 0.046
λ_{ma}	**0.0020**	$\mathbf{3.17 \pm 0.039}$
λ_{ma}	0.0050	3.28 ± 0.053
α	0	3.27 ± 0.051
α	0.0005	3.21 ± 0.044
α	**0.0010**	$\mathbf{3.17 \pm 0.039}$
α	0.0020	3.20 ± 0.043
α	0.0050	3.30 ± 0.057

The three bold rows represent the same final configuration, $\lambda_{\text{gwd}} = 0.05$, $\lambda_{\text{ma}} = 0.002$, and $\alpha = 0.001$; they therefore share the same result.

The cross-distance experiment is supported by the supplied 720-sample result sheet. The observed range was 1.848–11.952 m, with 232, 251, and 237 samples in the near, medium, and far intervals. Enabling $\mathcal{L}_{\text{ma}}$ reduced MAE from 3.680 to 3.040 mm in the near interval, from 3.950 to 3.130 mm in the medium interval, and from 4.360 to 3.340 mm in the far interval. Across all 720 samples, MAE decreased from 3.998 to 3.170 mm (20.71%), and RMSE decreased from 4.186 to 3.180 mm

(24.03%). Absolute error still increased with distance, so we claim consistent improvement over the evaluated range rather than distance invariance.

Stability of the measurement-aware loss across camera-to-target distances. Error entries are reported as $\mathcal{L}_{\mathrm{ma}}$ OFF/ON.

Distance group	Observed distance / m	Samples	MAE / mm	RMSE / mm	MAE reduction / mm	MAE reduction / %	RMSE reduction / %
Near	1.848–5.216	232	3.680/3.040	3.820/3.050	0.640	17.39	20.16
Medium	5.216–8.584	251	3.950/3.130	4.120/3.140	0.820	20.76	23.79
Far	8.584–11.952	237	4.360/3.340	4.580/3.343	1.020	23.39	27.01
Overall	1.848–11.952	720	3.998/3.170	4.186/3.180	0.828	20.71	24.03

Within the tested ranges, the final configuration produced the lowest mean MAE. Moving away from the selected values increased MAE, but the changes between neighboring settings were generally gradual. This indicates that the selected coefficients lie within a relatively stable operating region rather than at an isolated narrow point. Because the one-factor-at-a-time analysis did not test every joint combination of the three coefficients, we describe the result as the best tested coordinate-wise configuration rather than as a global optimum.

The standard YOLOv5 training hyperparameters were searched using its hyperparameter-evolution strategy.

The newly introduced λ_{gwd}, λ_{ma}, and α were determined separately by one-factor-at-a-time sensitivity analysis. When one coefficient was scanned, the other two were fixed at their final selected values, and each setting was repeated three times. The selected values of 0.05, 0.002, and 0.001, respectively, produced the lowest mean measurement error among the tested settings, while neighboring values showed only gradual changes.

Because the physical consequence of an image-domain error depends on camera-to-target distance, we divided 720 samples spanning 1.848–11.952 m into near, medium, and far groups of 232, 251, and 237 samples. Under otherwise identical conditions, $\mathcal{L}_{\mathrm{ma}}$ reduced MAE from 3.680 to 3.040 mm in the near group and from 4.360 to 3.340 mm in the far group. Overall MAE decreased from 3.998 to 3.170 mm and RMSE from 4.186 to 3.180 mm. Improvement in all three groups indicates that the loss remained effective as imaging distance changed.

R1.Q8 GCER depends on local edge responses under occlusion and dust, but its retained boundary point quality is evaluated only through aggregate MAE (p. 8).

Response We thank the reviewer for identifying an important gap in the evidence used to evaluate GCER. We agree that the aggregate physical-size MAE reported in the original manuscript showed only the downstream measurement effect of refinement. It did not directly establish whether the local boundary observations retained after outlier rejection were closer to the manually corrected contour. We have therefore added a point-level boundary-quality audit and module-level diagnostics on field images with different levels of coal-dust interference and physical occlusion.

In the new evaluation, the detector outputs and GCER parameters were held fixed. For each initial ellipse, GCER uniformly sampled 96 contour locations and searched for the maximum local gradient

response along the normal direction within a radius of $R_s = 4$ pixels. The selected observations were screened using the median absolute deviation (MAD) of their normal offsets, and at least 24 reliable points were required for ellipse refitting. If the joint offset and edge-response criteria retained too few points, the edge-response criterion was relaxed; if fewer than 24 MAD-consistent observations remained, the refinement was rejected and the initial ellipse was returned. This description has also been added to Section II-D and Algorithm 1 so that the manuscript now states the implemented fallback behavior explicitly.

For direct point-level assessment, we computed the nearest Euclidean distance from every candidate or retained observation to the manually corrected reference contour. We report the mean, median, and 95th-percentile distances, together with the fraction of points whose distance was within 0.5% or 1% of the corresponding reference major-axis diameter. The point-level audit covered 324 roller tank lug instances from 200 real mine images. GCER reduced the mean distance from 2.643 to 2.425 pixels (8.25%), the median distance from 1.691 to 1.473 pixels (12.89%), and the P95 distance from 7.141 to 6.876 pixels (3.71%). The fractions of points within the 0.5% and 1% diameter tolerances increased from 60.63% to 66.34% and from 76.94% to 82.34%, respectively. Ellipse IoU increased from 0.9616 to 0.9706; it improved in 287 instances, decreased in 35, and remained essentially unchanged in two. The smaller P95 improvement indicates that GCER mainly corrected the central part of the boundary-error distribution and observations close to the tolerance thresholds, while a small number of large deviations remained.

We further evaluated GCER on 180 field images affected by coal dust and 240 field images containing physical occlusion. The coal-dust set contained 60 mild, 60 moderate, and 60 severe cases, while the physical-occlusion set contained 80 cases at each severity level. For every image, the detector output, GCER parameters, and manually corrected reference contour were fixed, and the same point-level metrics were calculated before and after refinement. The "accepted/fallback" count indicates whether sufficient MAD-consistent observations remained for ellipse refitting; acceptance does not itself imply that the retained responses belong to the physical boundary.

Under mild and moderate coal dust, the mean distance decreased from 2.38 to 2.32 pixels and from 2.70 to 2.63 pixels, respectively, while ellipse IoU increased from 0.968 to 0.972 and from 0.948 to 0.953. Mild and moderate physical occlusion showed the same direction of change: the mean distance decreased from 2.78 to 2.68 pixels and from 3.32 to 3.20 pixels, and IoU increased from 0.949 to 0.959 and from 0.914 to 0.926. These changes are small but consistent across the mean distance, P95 distance, 1%-diameter tolerance, and IoU.

GCER did not continue to improve the boundary under severe interference. Under severe coal dust, the mean distance changed from 3.42 to 3.43 pixels and IoU from 0.909 to 0.907; under severe physical occlusion, the corresponding values changed from 4.62 to 4.63 pixels and from 0.846 to 0.843. Strong dust can produce coherent gradients that compete with the physical boundary, while extensive occlusion removes much of the contour needed for refitting. The fallback was triggered in one severe-dust image and eight severe-occlusion images. It prevents unstable fitting when too few

reliable observations remain, but it cannot recover boundary information absent from the image. We therefore limit the GCER claim to conditions in which sufficient physical contour remains visible and do not claim reliable correction under severe interference.

To determine whether GCER retained points closer to the physical boundary, we measured their nearest distance to the manually corrected contour and the fractions falling within 0.5% and 1% of the reference major-axis diameter. We further evaluated GCER under different levels of coal-dust interference and physical occlusion.

GCER used $K = 96$ uniformly spaced contour samples, a normal search radius of $R_s = 4$ pixels, and an MAD rejection coefficient of $\eta = 1.8$; an ellipse was refitted when at least $N_{\min} = 24$ observations remained. This configuration was fixed in all GCER experiments.

Condition	Level	Images	Mean / px	P95 / px	$P_{1\%}$ / %	IoU	Accepted / fallback
Coal dust	Mild	60	2.38→2.32	6.48→6.31	78.3→79.1	0.968→0.972	60 / 0
Coal dust	Moderate	60	2.70→2.63	7.10→6.92	72.1→73.0	0.948→0.953	60 / 0
Coal dust	Severe	60	3.42→3.43	9.15→9.18	60.6→60.4	0.909→0.907	59 / 1
Physical occlusion	Mild	80	2.78→2.68	7.00→6.78	46.0→47.6	0.949→0.959	80 / 0
Physical occlusion	Moderate	80	3.32→3.20	8.35→8.12	37.0→38.6	0.914→0.926	79 / 1
Physical occlusion	Severe	80	4.62→4.63	11.55→11.60	25.5→25.3	0.846→0.843	72 / 8

The general point-level audit covered 324 lug instances from 200 images. GCER reduced the mean and median point-to-contour distances from 2.643 to 2.425 pixels and from 1.691 to 1.473 pixels, respectively, while the P95 distance decreased from 7.141 to 6.876 pixels. The fractions within 0.5% and 1% of the reference major-axis diameter increased from 60.63% to 66.34% and from 76.94% to 82.34%, and ellipse IoU increased from 0.9616 to 0.9706.

GCER improved the mean distance, P95 distance, and ellipse IoU under mild and moderate coal dust or physical occlusion. Under severe interference, these metrics remained nearly unchanged or deteriorated slightly. Strong dust edges may be selected as local responses, while extensive physical occlusion removes much of the true contour. GCER can refine the initial ellipse when sufficient physical boundary remains visible, but it cannot reconstruct boundary information that is absent from the image.

R1.Q9 The multi frame fusion evidence comes from one thirty frame window at a single actual lug size, so the robustness claim is narrow.

Response We thank the reviewer for identifying that the original presentation did not distinguish clearly between the representative temporal visualization and the larger-scale evaluation of multi-frame fusion. We agree that the original wording, which introduced a single 30-frame window corresponding to a 370-mm lug immediately before the fusion results, could reasonably be interpreted as implying that this window was the only evidence used to support the fusion claim. To remove this ambiguity, we deleted the representative temporal plot from the revised manuscript and reorganized the subsection around the aggregate evaluation.

The 720 representative test images were obtained from 720 different source videos, with one image retained from each video for image-level evaluation. These images contained 1147 roller tank lug target instances, all of which were used for object-detection evaluation. For physical-size measurement and cross-distance evaluation, one lug with a corresponding physical reference was selected from each test image, yielding 720 independent single-frame measurement samples. For the large-scale wear-state evaluation, 1000 target instances were selected to retain substantial representation of all three wear states: 350 normal, 350 slight-wear, and 300 severe-wear instances. Each selected instance was mapped back to its source video, and 30 consecutive frames containing the same lug were retrieved. The large-scale evaluation therefore comprised 1000 target-level 30-frame windows and 30,000 frame-level target measurements. A source video could contain more than one lug, so the number of target-level windows could exceed the number of source videos.

The single-frame MAE of 3.17 mm was evaluated on the 720 independent measurement samples described above. In the temporal comparison, one frame was randomly selected from each of the 1000 windows before fusion, producing the single-frame baseline MAE of 3.23 mm; the fusion methods used all 30 frames from the same windows. The single 370-mm sequence in the original manuscript was used only to illustrate temporal behavior and was not the source of the reported quantitative results. The reduction in MAE from 3.23 to 2.63 mm was computed over all 1000 target-level 30-frame windows. The same 1000-window evaluation showed that adding multi-frame fusion increased accuracy from 97.50% to 99.20%, Macro-F1 from 97.43% to 99.20%, and severe-wear recall from 96.71% to 99.33%.

We have also bounded the robustness statement. The revised manuscript states that the results support improved final wear-state assessment over the evaluated videos, target-level windows, and wear conditions. They do not establish failure-free fusion under prolonged complete occlusion or under a persistent systematic measurement bias shared by all frames in a window.

The multi-frame experiment used 1000 target-level 30-frame windows, comprising 30,000 frame-level measurements. For the single-frame baseline, one frame was randomly selected from each window before temporal processing, yielding one unfused measurement per target window. Mean fusion, median fusion, Kalman filtering, and the proposed robust weighted fusion used all 30 frames from the same windows.

The wear-state experiment used 1000 target-level 30-frame windows selected from the 1147 test instances: 350 normal, 350 slight-wear, and 300 severe-wear windows. Each window follows the same lug through 30 consecutive frames and produces one fused size and one wear state. Because one video may contain several lugs, the number of target windows can exceed the number of videos. The experiment comprised 30,000 frame-level measurements.

Across 1000 target-level 30-frame windows, robust multi-frame fusion reduced the overall MAE from 3.23 to 2.63 mm, and the final pipeline achieved an accuracy of 99.20%, a macro-averaged F1 score (Macro-F1) of 99.20%, and a severe-wear recall of 99.33%.

R1.Q10 The final wear classification uses balanced one thousand samples, without a confusion matrix or false negative counts near the four mm boundary (p. 11).

Response We thank the reviewer for identifying this important weakness in the original evaluation. We fully agree that the aggregate results on 1000 test samples did not adequately characterize decision reliability near the maintenance thresholds. After re-examining the original test set, we found that relatively few samples were concentrated immediately around the 4-mm warning threshold or the 8-mm replacement threshold. Consequently, the original aggregate accuracy could obscure errors caused by small measurement deviations near a hard decision boundary.

We therefore added an independent boundary-focused evaluation containing 400 samples. Four intervals of the wear-related dimensional-deviation indicator were evaluated: 3–< 4 mm, 4–5 mm, 7–< 8 mm, and 8–9 mm, with 100 samples in each interval. The first two intervals assess the 4-mm warning boundary, whereas the latter two assess the 8-mm replacement boundary. Unlike the original full-distribution evaluation, every sample in this analysis lies within 1 mm on either side of the corresponding threshold.

We also added the aggregate confusion matrix requested by the reviewer for the original 1000-window evaluation. The evaluation set comprised 350 normal, 350 slight-wear, and 300 severe-wear windows. The final method correctly classified 347, 347, and 298 windows in these three ground-truth classes, respectively. The resulting accuracy was 99.20%, Macro-F1 was 99.20%, and severe-wear recall was 99.33%. Two severe-wear windows were assigned to slight wear, and none was assigned directly to the normal class. The confusion matrix is now reported as a separate figure so that the aggregate class-specific errors can be inspected independently of the boundary-focused experiment.

At the 4-mm boundary, 73 of the 100 samples in the 3–< 4 mm interval were classified as normal, while 24 and three were conservatively classified as slight and severe wear, respectively. Of the 100 samples in the 4–5 mm interval, 89 were correctly classified as slight wear, two were misclassified as normal, and nine were classified as severe wear. The resulting boundary accuracy was 81.0%, and the false-negative rate for slight wear at the 4-mm boundary was 2.0% (2/100).

At the 8-mm boundary, 78 of the 100 samples in the 7–< 8 mm interval were correctly classified as slight wear, while four and 18 were classified as normal and severe wear, respectively. Of the 100 samples in the 8–9 mm interval, 97 were correctly classified as severe wear, two were classified as slight wear, and one was classified as normal. The resulting boundary accuracy was 87.5%, and severe-wear recall was 97.0%; equivalently, three severe-wear samples were classified as non-severe.

These boundary results are lower than the 99.20% aggregate accuracy because the two evaluations address different operating regimes. The original test set represents the full distribution and contains many samples well separated from either threshold. In contrast, the added evaluation deliberately concentrates all samples within 1 mm on either side of a hard threshold. In the proposed pipeline, wear state is not produced by an independent classifier. Continuous ellipse geometry is converted into physical dimensions and then into a wear-related dimensional-deviation indicator

before threshold assignment. A residual millimeter-level geometric error can therefore change the discrete state of a sample located close to 4 or 8 mm even when the target remains correctly detected.

The boundary errors can arise in both directions from the initial ellipse, the finite GCER search range, edge visibility, and subsequent camera-based size reconstruction. GCER searches for local edge responses within a finite normal-direction neighborhood of the initial ellipse. If the initial ellipse is undersized and the true outer boundary is weak, occluded, or outside the effective search neighborhood, retained points may remain inside the physical contour. Refitting then underestimates the lug diameter and overestimates wear, which can move a normal boundary sample toward a higher wear category. Returning the initial ellipse when too few reliable points remain prevents unstable refitting but does not remove this inward bias.

Conversely, if the initial ellipse is oversized, the true boundary may again lie outside the effective search neighborhood, or local background structures, coal-dust contours, and other strong gradients may be selected instead of the physical boundary. GCER may then reduce the initial error without pulling every sampled point precisely onto the true contour, leaving a residual outward bias in the refitted ellipse. For a worn lug whose true dimensions are below the nominal diameter, such size overestimation can reduce the estimated wear and may move a slight-wear sample toward the normal category. This is a possible explanation for the two false-negative cases in the 4–5 mm interval.

GCER corrects local contour positions in the image domain but cannot remove uncertainty introduced during physical-size reconstruction. From $W = 2da/f_x$ and $H = 2db/f_y$, the first-order relative error propagation is approximately

$$\frac{\Delta W}{W} \approx \frac{\Delta a}{a} + \frac{\Delta d}{d} - \frac{\Delta f_x}{f_x}, \qquad \frac{\Delta H}{H} \approx \frac{\Delta b}{b} + \frac{\Delta d}{d} - \frac{\Delta f_y}{f_y}.$$

Residual uncertainty in the focal lengths, camera-to-target distance, and ellipse semi-axes therefore propagates to the reconstructed dimensions. In the off-axis correction, residual principal-point errors in c_x and c_y additionally affect the effective viewing angle. Consequently, a locally accurate GCER contour does not guarantee an error-free millimeter-level reconstruction. Boundary-neighborhood errors should therefore be interpreted as the combined propagation of initial ellipse error, finite local refinement, edge visibility, camera calibration, and distance measurement, rather than as a failure of a single module. Although conservative assignments to higher wear categories predominated, the observed false negatives show that the method does not guarantee zero under-reporting.

We have added the boundary confusion counts and this interpretation to the revised manuscript. We now explicitly distinguish full-distribution performance from threshold-neighborhood performance and acknowledge that the current hard-threshold decision is sensitive to residual geometric uncertainty. In field operation, a result close to either maintenance threshold should prompt con-

firmation from additional frames and manual inspection rather than an irreversible maintenance decision based on a single numerical estimate. Reducing conservative false alarms while preserving low safety-critical false-negative rates will require more precise uncertainty-aware boundary measurement and is identified as future work.

Because the original 1000-window set contained few samples close to the 4- and 8-mm decision boundaries, we added 400 boundary samples: 100 in each of the 3–< 4, 4–5, 7–< 8, and 8–9 mm intervals. Every sample was within 1 mm on either side of its corresponding threshold.

The 4- and 8-mm subsets achieved accuracies of 81.0% and 87.5%, respectively. Two slight-wear samples were classified as normal, and three severe-wear samples as non-severe, giving a severe-wear recall of 97.0%. All samples were within 1 mm on either side of the corresponding threshold, whereas the system-level size error was approximately 2–3 mm; a small ellipse offset can therefore change the discrete state. If the outer contour is weak, occluded, or outside the 4-pixel GCER search range, the search may select an inner edge or nearby interference, making the reconstructed size too large or too small. Calibration and distance errors add further variation. Overestimation of wear was more common, reducing missed warnings but increasing false alarms near the thresholds. Measurements close to 4 or 8 mm should therefore be checked using additional frames and manual inspection.

R1.Q11 Reported real time performance is based on an RTX A6000 workstation, and end device latency for refinement plus fusion is not shown.

Response We thank the reviewer for identifying that the original manuscript did not describe the deployment architecture or report the computational cost of temporal fusion. We agree that referring only to an "RTX A6000 workstation" could suggest that the timing experiment was disconnected from the actual field implementation. We have therefore clarified the acquisition–processing architecture and added the refinement and fusion latency.

The system uses centralized server-side processing rather than camera-side embedded inference. Cameras installed at surface and underground monitoring positions, with the current operation relying primarily on the surface-side observations, transmit captured images to the mine server through Ethernet and optical-fiber links. The server then performs ellipse detection, GCER, physical-size reconstruction, temporal fusion, and wear-state assessment. The deployed field server is equipped with an Intel Xeon Silver 4210R CPU and an NVIDIA RTX A6000 GPU. The experiments and runtime measurements reported in this manuscript were performed directly on this mine-site server.

With a 640×640 input and a batch size of 1, the complete single-frame pipeline, including image preprocessing, ellipse detection and decoding, GCER, and physical-size reconstruction, achieved 35.35 ± 0.88 FPS. This corresponds to an average single-frame processing latency of 28.29 ms. The 30-frame fusion operation, including MAD-based outlier rejection, adaptive-weight calculation, and weighted aggregation, required 0.16 ms per window. Because fusion is executed once for each 30-frame window, its amortized cost is approximately 0.005 ms per frame, giving an average server-side

computational cost of 28.30 ms per frame. After the final frame of a window has been processed, fusion adds 0.16 ms, so the computation from arrival of the final frame to the fused output is approximately 28.45 ms.

Training, testing, and field inference were performed directly on the mine-site workstation, equipped with an Intel Xeon Silver 4210R CPU at 2.40 GHz and an NVIDIA RTX A6000 GPU under Windows 10. Cameras at the surface and underground monitoring positions transmit images to this workstation through Ethernet and optical-fiber links for centralized processing.

Runtime was measured with a 640×640 input and a batch size of 1 over five repeated runs, including the post-processing required by each method.

The complete method achieved 35.35 ± 0.88 server-side end-to-end FPS, or 28.29 ms per frame. MAD rejection and adaptive weighted fusion added only 0.16 ms per 30-frame window, giving an amortized server-side cost of 28.30 ms per frame and approximately 28.45 ms from arrival of the final frame to the fused output.

R1.Q12 More recent work to consider: "Enhanced Precision in Dam Crack Width Measurement: Leveraging Advanced Lightweight Network Identification for Pixel-Level Accuracy."

Response We thank the reviewer for recommending this recent study. It usefully broadens the discussion from category-level industrial detection to measurement-oriented pixel localization. We have read and cited the work by Wu et al., which combines a lightweight MobileNetV2–DeepLabV3+ segmentation network, maximum-inscribed-circle estimation, and depth-camera geometry to convert crack pixels into physical crack width.

The revised Introduction uses this work to clarify both the common objective and the methodological distinction. Both studies recognize that image-domain localization errors propagate into physical measurement. The dam-crack method estimates the width of an irregular slender region after semantic segmentation, whereas our method predicts a global ellipse representation for a roller tank lug and reconstructs its major and minor dimensions under partial boundary visibility. We have therefore cited the paper as a complementary example of learning-assisted geometric measurement, rather than using it as a generic object-detection reference.

In machine-vision-based dimensional measurement, a common method is to extract the target boundary from the image and then calculate its physical dimensions using imaging geometry. Luo et al. measured grape berry size using edge detection and contour reorganization, while Wu et al. combined crack segmentation, maximum-inscribed-circle estimation, and depth-camera geometry to calculate crack width. These studies demonstrate that edge and contour geometry provide a clear and interpretable basis for physical-size measurement. Under coal dust, background clutter, or partial occlusion, however, target boundaries may become fragmented or displaced, which subsequently affects contour fitting and size reconstruction. Considering these characteristics of field images of roller tank lugs, the present framework uses a learned global ellipse to represent the overall target shape and then combines local boundary refinement with a camera model for

physical-size measurement.

Response to Reviewer 2

R2.General This paper presents a measurement-aware ellipse detection framework for real-time lug wear assessment. This work has certain engineering application value, but some issues still require further clarification.

Response We thank the reviewer for recognizing the engineering value of the proposed framework and for identifying aspects that required further clarification. We now distinguish the learned detector from the deterministic metrology, refinement, fusion, and threshold-based assessment stages. We also explain the physical meaning and implementation of the measurement-aware objective and provide additional evidence on calibration sensitivity, projection correction, temporal filtering, runtime cost, and the use of YOLOv5 as the implementation backbone. Finally, we limit the conclusions to the evaluated near-frontal acquisition conditions. Our point-by-point responses and the corresponding manuscript revisions are provided below.

R2.Q1 What is the specific method for ellipse annotation in the self-built dataset? How are the physical dimensions Wgt, Hgt, and camera distance d obtained?

Response We thank the reviewer for requesting clarification of the ellipse annotation procedure and the acquisition of W_{gt}, H_{gt}, and d. The image-domain ellipse annotations and the physical reference measurements were obtained through independent procedures.

For ellipse annotation, the roller tank lug contours were first annotated in CVAT and exported as segmentation polygons. The normalized polygon coordinates were converted to image coordinates, and OpenCV `fitEllipse` was used to obtain an initial ellipse parameterized by its center, semi-major axis, semi-minor axis, and orientation. Local edge information was subsequently used to assist boundary adjustment. Importantly, the automatically initialized and edge-adjusted ellipses were not directly accepted as ground truth. Every annotation was manually inspected on an image-by-image basis, and its center, semi-axes, and orientation were corrected to ensure close agreement with the visible outer boundary. For partially occluded targets, ellipse continuity was checked against the remaining visible contour. Only annotations that passed this manual quality-control procedure were retained in the final dataset.

The physical dimensions W_{gt} and H_{gt} were obtained from repeated field measurements rather than inferred from the image annotations. Three field staff members each performed ten measurements. A rigid plate was placed against one side of the roller tank lug as a reference surface, laser range measurements were taken from the opposite side, and the resulting dimensions were cross-checked with a large-range industrial vernier caliper. The camera-to-target distance d was measured by the laser distance meter at the corresponding acquisition position and stored with the associated images. The corresponding annotation and measurement procedures have been added to Section IV-B.

Ellipse annotations were initialized from CVAT polygon contours using OpenCV `fitEllipse`, based

on the direct least-squares ellipse-fitting method of Fitzgibbon et al. Each result was then checked against the visible physical boundary and manually corrected when necessary. Partially occluded samples were assessed from the remaining visible contour.

R2.Q2 Formulas (8) and (14) are based on a pinhole camera model, assuming the lug is located near the optical axis and its orientation is approximately facing the camera. However, actual underground imaging may have a large skew angle. Although formulas (15)–(18) introduce side projection correction, they do not specify the effective skew angle range, nor do they provide experimental verification of the correction effect.

Response We thank the reviewer for requesting an explicit operating range and direct verification. We agree that the original manuscript did not clearly distinguish lateral off-axis displacement from arbitrary three-dimensional skew. In the actual installation, the camera and roller tank lug were arranged under a near-frontal acquisition constraint, so the lug plane remained approximately parallel to the image plane. The side-projection model was designed to correct the residual horizontal and vertical displacement of the lug center within the field of view. It was not intended to solve unrestricted out-of-plane lug inclination.

For substantially non-orthogonal poses, Lu et al. reported a camera–LiDAR approach that combines YOLOv8-based localization, camera–LiDAR joint calibration, image–point-cloud fusion, and inclination-adaptive geometric reconstruction [X. Lu, Y. Shen, H. Feng, X. Bai, H. Hao, and J. Song, "Research on a real-time dimensions detection method for roller tank lugs in non-orthogonal attitude within vertical shaft hoisting systems," *Measurement Science and Technology*, vol. 37, no. 4, Art. no. 045016, 2026, doi: 10.1088/1361-6501/ae3793]. That method addresses dimensional reconstruction under pronounced three-dimensional non-orthogonal attitudes. By contrast, the present monocular method addresses mechanically constrained near-frontal acquisition with clutter, partial occlusion, incomplete boundaries, and residual target-center displacement from the optical axis.

To make this distinction quantitative, we now define the effective off-axis angle from the target-center viewing ray:

$$\beta = \arctan\left(\frac{\sqrt{m_x^2 + m_y^2}}{d}\right), \qquad m_x = \frac{d(u - c_x)}{f_x}, \quad m_y = \frac{d(v - c_y)}{f_y}.$$

This angle is formed by the target-center viewing ray and the camera optical axis. It is therefore different from the in-plane ellipse orientation θ and from a three-dimensional normal-vector tilt angle. We further performed a paired ablation on 50 samples of the 360-mm lug specification, spanning an observed β range of 0.307°–5.335°. Only the side-projection correction was switched, while the detector output, camera parameters, recorded distance, and ground truth were held fixed.

The correction reduced overall MAE from 5.169 to 3.347 mm, RMSE from 5.770 to 3.371 mm, and

MAPE from 1.436% to 0.930%. The average error decreased in all four off-axis-angle groups, and the MAE reduction increased from 10.63% for Q1 (0.307°–1.280°) to 48.30% for Q4 (3.561°–5.335°). These results show that the effect of the correction becomes more pronounced as the target center moves farther from the optical axis and provide direct experimental support within the observed near-frontal range.

Paired evaluation of side-projection correction. Error entries are reported as correction OFF/ON.

Group	Observed β / deg	n	MAE / mm	RMSE / mm	MAPE / %	MAE reduction / %
Q1	0.307–1.280	13	3.537/3.161	3.738/3.171	0.982/0.878	10.63
Q2	1.406–2.563	12	4.978/3.356	5.282/3.373	1.383/0.932	32.57
Q3	2.579–3.352	12	5.451/3.405	6.039/3.432	1.514/0.946	37.54
Q4	3.561–5.335	13	6.718/3.473	7.394/3.504	1.866/0.965	48.30
Overall	0.307–5.335	50	5.169/3.347	5.770/3.371	1.436/0.930	35.24

The field cameras were installed at different monitoring positions for near-frontal imaging, with the lug plane approximately parallel to the image plane of the corresponding camera. The correction therefore addresses the side-projection error caused by displacement of the target center from the optical axis.

A paired correction-on/off test used 50 near-frontal samples with target-center viewing angles of 0.307°–5.335°. With identical detections, camera parameters, distances, and physical references, the side-projection correction reduced overall MAE from 5.169 to 3.347 mm.

R2.Q3 Are the focal lengths fx and fy pre-calibrated? Do they change with camera zoom or replacement?

Response Yes. Multiple fixed-focus cameras were used at different monitoring positions and heights. Each camera, lens, and resolution configuration was calibrated separately using Zhang's method before acquisition. Neither optical nor digital zoom was applied, and the focal length and image resolution of each camera remained fixed during its corresponding acquisition sessions. Each sample was reconstructed using the intrinsic parameters of its source camera. These parameters are not assumed to remain valid after replacement of the camera or a change in the lens, resolution, or imaging configuration; in any of these cases, the affected camera must be recalibrated before physical-size reconstruction. We have added this clarification to Section IV-A (Experimental Setup).

Each camera, lens, and resolution configuration was calibrated separately using Zhang's method. The focal length and image resolution of each camera remained fixed during its acquisition sessions, and each sample was reconstructed using the intrinsics of its corresponding source camera.

R2.Q4 In Algorithm 1, the maximum gradient point is taken within a fixed search radius ρ in the normal direction. However, in cases of severe occlusion or extremely low illumination, the maximum

gradient point may originate from the background texture rather than the true boundary. In this case, can the anomaly removal mechanism (median absolute deviation) effectively distinguish between them?

Response We thank the reviewer for identifying this important limitation of local maximum-gradient search. We agree that MAD is not a semantic classifier and therefore cannot determine directly whether an edge response belongs to the roller tank lug, an occluder, or the background. Its role is narrower: it tests whether the normal offset of an observed edge is consistent with the dominant displacement estimated from the other sampled locations. Consequently, an isolated background response with an abnormal offset can often be rejected, but a spatially coherent occluder or background edge may remain if its offset is similar to that of the true contour.

The implemented GCER procedure reduces this risk through four safeguards. First, the normal search is restricted to the local interval $[-R_s, R_s]$, with $R_s = 4$ pixels in the reported GCER audits. Second, candidate observations are ranked by their edge responses after the maximum-gradient selection. Third, MAD rejects observations with inconsistent normal offsets. Fourth, at least $N_{\min} = 24$ reliable observations are required for refitting. If the joint offset and edge-response selection leaves too few points, the edge-response restriction is relaxed and the MAD-consistent set is used; if fewer than 24 MAD-consistent observations remain, the refinement is rejected and the initial ellipse is returned. This fallback prevents an unstable ellipse from being fitted to a very small point set, but it does not recover a contour that is largely invisible.

We quantified this behavior on field images with different levels of coal-dust interference and physical occlusion. Under mild and moderate coal dust, the mean point-to-contour distance decreased from 2.38 to 2.32 pixels and from 2.70 to 2.63 pixels, respectively. Under mild and moderate physical occlusion, it decreased from 2.78 to 2.68 pixels and from 3.32 to 3.20 pixels. The corresponding P95, 1%-diameter tolerance, and IoU values changed in the same favorable direction. Under severe coal dust and severe occlusion, however, the mean distance changed from 3.42 to 3.43 pixels and from 4.62 to 4.63 pixels, while IoU decreased slightly from 0.909 to 0.907 and from 0.846 to 0.843. Thus, MAD-based offset consistency is useful when enough physical contour remains visible, but it cannot distinguish a coherent false edge from the true boundary under severe interference.

GCER point-level and robustness diagnostics

Condition	Level	Images	Mean / px	P95 / px	$P_{1\%}$ / %	IoU	Accepted / fallback
Coal dust	Mild	60	2.38→2.32	6.48→6.31	78.3→79.1	0.968→0.972	60 / 0
Coal dust	Moderate	60	2.70→2.63	7.10→6.92	72.1→73.0	0.948→0.953	60 / 0
Coal dust	Severe	60	3.42→3.43	9.15→9.18	60.6→60.4	0.909→0.907	59 / 1
Physical occlusion	Mild	80	2.78→2.68	7.00→6.78	46.0→47.6	0.949→0.959	80 / 0
Physical occlusion	Moderate	80	3.32→3.20	8.35→8.12	37.0→38.6	0.914→0.926	79 / 1
Physical occlusion	Severe	80	4.62→4.63	11.55→11.60	25.5→25.3	0.846→0.843	72 / 8

We have therefore revised both the mechanism description and the scope of the claim. GCER is presented as effective under mild-to-moderate occlusion when sufficient distributed contour evidence remains. We do not claim that MAD can reliably distinguish true and background edges under

severe occlusion, extremely low illumination, or extensive contour loss. In these cases, coherent false edges may pass the geometric consistency test, or too few reliable points may remain; the system then either returns the initial ellipse or may retain residual boundary error. This limitation is now stated explicitly in Sections II-D and IV-D.

MAD does not semantically classify an edge response as belonging to the lug, an occluder, or the background. Instead, it rejects observations whose normal offsets are inconsistent with the dominant boundary displacement. Isolated background responses can therefore often be removed under mild-to-moderate occlusion, whereas spatially coherent occluder or background edges may remain when their offsets resemble the true contour. Returning E_0 when fewer than $N_{\min}$ reliable observations remain prevents unstable refitting but cannot reconstruct a largely invisible contour. GCER is consequently not expected to provide reliable correction under severe or extensive occlusion.

GCER improved the mean distance, P95 distance, and ellipse IoU under mild and moderate coal dust or physical occlusion. Under severe interference, these metrics remained nearly unchanged or deteriorated slightly. Strong dust edges may be selected as local responses, while extensive physical occlusion removes much of the true contour. GCER can refine the initial ellipse when sufficient physical boundary remains visible, but it cannot reconstruct boundary information that is absent from the image.

R2.Q5 The weights $\gamma_i = q_i r_i u_i$ are in a product form, but the dimensions and distributions of the three components are different; the rationale for direct multiplication needs further explanation.

Response We thank the reviewer for identifying that the dimensional consistency and design rationale of the adaptive fusion weight were insufficiently explained. We agree that the original presentation could be interpreted as multiplying three raw quantities with different physical units. We have therefore clarified that q_i, r_i, and u_i are dimensionless reliability scores representing detection confidence, median consistency, and temporal stability, respectively.

The product $\gamma_i = q_i r_i u_i$ is an engineering soft gate rather than a product of statistically independent probabilities. Here, $q_i = \hat{c}_{\mathrm{det},i}$ is dimensionless; $r_i = \exp[-\Delta_i^{\mathrm{med}}/(\mathrm{MAD} + \epsilon)]$ normalizes the deviation from the window median by a robust scale with the same units; and $u_i = \exp[-\Delta_i^{\mathrm{temp}}/(\sigma_s + \epsilon)]$ normalizes the adjacent-frame size difference by the window-adaptive temporal scale σ_s. Consequently, all three factors lie in $(0, 1]$. A low value in any factor suppresses that frame rather than being compensated by large values in the other factors. Before weighting, MAD-based screening rejects measurements with excessive robust deviation. The normalized coefficient $\widetilde{\gamma}_i = \gamma_i / \sum_{j\in\Omega} \gamma_j$ makes the fused result depend on relative frame reliability rather than the absolute scale of the product. We do not claim that this weighting function is theoretically unique or globally optimal.

The fusion weight is assigned adaptively as

$$\gamma_i = q_i \, r_i \, u_i,$$

where q_i, r_i, and u_i represent detection confidence, median consistency, and temporal stability, respectively. Specifically,

$$q_i = \hat{c}_{\mathrm{det},i},$$

where $\hat{c}_{\mathrm{det},i}$ is the single-class detection confidence of frame i and is used only as a dimensionless frame-reliability score. The median-consistency weight is

$$r_i = \exp\left(-\frac{\Delta_i^{\mathrm{med}}}{\mathrm{MAD} + \epsilon}\right),$$

which assigns lower weights to measurements far from the local median. The temporal-stability weight is

$$u_i = \exp\left(-\frac{\Delta_i^{\mathrm{temp}}}{\sigma_s + \epsilon}\right),$$

where

$$\Delta_i^{\mathrm{temp}} = \|S_i - S_{i-1}\|_2,$$

and the window-adaptive temporal scale is

$$\sigma_s = \mathrm{median}\left(\left\{\|S_j - S_{j-1}\|_2\right\}_{j=2}^{N}\right).$$

For the first frame in the temporal window, u_1 is set to 1. All three factors are dimensionless reliability scores in $(0, 1]$. The confidence q_i is already dimensionless, whereas r_i and u_i normalize size deviations by window-adaptive scales with the same units. Their product is therefore a conjunctive soft gate rather than a multiplication of heterogeneous physical quantities.

R2.Q6 The temporal stability weight u_i depends on the difference between adjacent frames, but when the target moves slowly, a small frame difference does not necessarily indicate accurate measurement and may introduce smoothing bias. It is recommended to compare it with other temporal filtering methods (such as Kalman filtering) to verify its superiority.

Response We thank the reviewer for this important and well-targeted comment. We fully agree that a small difference between adjacent frames does not establish measurement accuracy and may preserve a shared systematic bias. In our formulation, $u_i = \exp[-\Delta_i^{\mathrm{temp}}/(\sigma_s + \epsilon)]$ is a temporal-stability weight, where $\Delta_i^{\mathrm{temp}} = \|S_i - S_{i-1}\|_2$. Its purpose is to suppress abrupt frame-to-frame measurement changes, not to certify that a slowly varying estimate is accurate. We have clarified this interpretation, added the requested Kalman-filter comparison, and bounded the resulting claim.

The temporal-stability term is therefore used only as one component of the frame-reliability weight. For a retained frame, $\gamma_i = q_i r_i u_i$ combines detection confidence, agreement with the robust window median, and resistance to abrupt temporal changes. Before weighting, MAD-based screening rejects measurements with excessive robust deviation. A temporally smooth frame can still receive a low contribution, or be rejected entirely, when its detection confidence is low or its reconstructed size

disagrees with the window median.

Following the reviewer's recommendation, we evaluated single-frame measurement, mean fusion, median fusion, Kalman filtering, and the proposed robust weighted fusion on the same 1000 target-level 30-frame windows. For the single-frame baseline, one frame was randomly selected from each window before fusion, yielding one unfused measurement per target window. Mean fusion, median fusion, Kalman filtering, and the proposed method received all 30 frame-level detections and reconstructed dimensions from the same windows; only the temporal processing strategy was changed. The comparison therefore involved 30,000 frame-level target measurements.

For the Kalman baseline, the state and observation vectors were defined in the physical-size domain as $\mathbf{x}_t = [W_t, H_t]^\top$ and $\mathbf{z}_t = [\hat{W}_t, \hat{H}_t]^\top$. Because the physical dimensions remain constant within a 30-frame window, we used the random-walk state-space model $\mathbf{x}_t = \mathbf{x}_{t-1} + \mathbf{w}_t$ and $\mathbf{z}_t = \mathbf{x}_t + \mathbf{v}_t$, corresponding to $\mathbf{F} = \mathbf{H} = \mathbf{I}_2$. The diagonal observation-noise covariance was estimated exclusively from all valid frame-level long- and short-axis residuals on the validation set:

$$\mathbf{R} = \mathrm{diag}(0.864, 0.756)\ \mathrm{mm}^2,$$

which corresponds to observation-noise standard deviations of 0.930 and 0.869 mm for the major and minor dimensions, respectively. The process-noise covariance was parameterized as $\mathbf{Q} = q_{\mathrm{K}}\mathbf{I}_2$. We searched

$$q_{\mathrm{K}} \in \{0, 10^{-4}, 3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}, 10^{-1}, 3\times10^{-1}, 1\}\ \mathrm{mm}^2$$

on the validation set. For each candidate, Kalman filtering was applied to every validation window, and the final posterior state of each 30-frame window was used as its fused estimate. The candidate minimizing the validation-set window-level MAE was selected, with RMSE used only to break a tie. This procedure yielded $q_{\mathrm{K}}^* = 0.030\ \mathrm{mm}^2$. Each sequence was initialized with $\mathbf{x}_{1|1} = \mathbf{z}_1$ and $\mathbf{P}_{1|1} = \mathbf{R}$. If a frame lacked a valid observation, only the prediction step was performed. The resulting $\mathbf{R}$ and q_{K}^* were fixed before testing; no Kalman parameter was selected or adjusted using test results.

Kalman filtering reduced the MAE from 3.230 mm for single-frame measurement to 2.680 mm, confirming that recursive temporal smoothing suppressed random frame-level fluctuations. The proposed robust weighted fusion achieved the lowest aggregate errors, with an MAE of 2.630 mm, an RMSE of 2.640 mm, and a MAPE of 0.710%. Relative to Kalman filtering, it reduced MAE and RMSE by 1.87% and 2.22%, respectively, while MAPE decreased by 0.012 percentage points. We do not describe this difference as statistically significant or universally superior because no inferential test was performed. The modest aggregate reduction indicates that robust rejection and confidence- and median-consistency weighting provided an additional benefit under the evaluated conditions.

The evaluated windows retained transient partial occlusion, coal-dust interference, local boundary

loss, and associated confidence degradation encountered in the field data. With a fixed observation-noise model, an occlusion-corrupted measurement can pull a Kalman estimate away from the true physical size and influence subsequent recursive updates. In the proposed method, such a frame can instead be rejected by the MAD criterion or downweighted through q_i and r_i before fusion. This mechanism is consistent with the lower aggregate errors observed for the proposed method. However, the 1000 windows were not further stratified by occlusion severity; we therefore do not claim a quantified advantage at any particular occlusion level.

We also explicitly acknowledge the reviewer's systematic-bias concern. If multiple consecutive frames share a persistent bias, both their median consistency and adjacent-frame temporal stability may remain high; consequently, the proposed weighting cannot identify that shared bias from temporal smoothness alone. Under this condition, neither the proposed fusion nor a standard Kalman filter is guaranteed to recover the true size. We have therefore limited our conclusion to the evaluated 1000 target-level windows: the proposed method achieved the lowest aggregate error, but it is not claimed to outperform Kalman filtering under every motion condition, occlusion level, or noise model.

For the Kalman baseline, the state and observation were the reconstructed major and minor dimensions, with identity transition and observation matrices. The validation residuals gave $\mathbf{R} = \mathrm{diag}(0.864, 0.756)\ \mathrm{mm}^2$. We set $\mathbf{Q} = q_{\mathrm{K}}\mathbf{I}_2$ and selected $q_{\mathrm{K}}^{*} = 0.030\ \mathrm{mm}^2$ by validation-window MAE. These parameters were fixed before testing.

The proposed method achieved the lowest MAE of 2.630 mm, with an RMSE of 2.640 mm and a MAPE of 0.710%. Its MAE and RMSE were 1.87% and 2.22% lower than those of Kalman filtering, although the difference was small. Short occlusion, coal dust, and local boundary loss occasionally corrupt individual frames. Such observations enter the Kalman recursion under a fixed noise model, whereas the proposed method rejects clear outliers by MAD and reduces the weights q_i and r_i of unstable frames. The two methods are therefore similar on stable frames, while the proposed fusion is less affected by transient unreliable measurements.

R2.Q7 The YOLO series has already evolved to YOLOv13. Why did the paper choose v5?

Response We thank the reviewer for this important question. We agree that the original manuscript did not explain sufficiently why YOLOv5 was retained when several later YOLO architectures are available. Our choice does not imply that YOLOv5 is superior to recent detectors in conventional object detection. Rather, YOLOv5s was selected because its mature and modular implementation allowed us to modify the prediction head while keeping the backbone and neck unchanged. This controlled design makes it possible to attribute performance changes to the ellipse representation, GWD, and measurement-aware supervision rather than to simultaneous changes in the underlying detector architecture.

Following the reviewer's suggestion, we added YOLOv8s, YOLOv10s, YOLO11s, YOLOv13-S, and RT-DETR-L, together with YOLOv8s-seg followed by `fitEllipse` as a segmentation-and-fitting

measurement baseline. All methods were evaluated in terms of conventional object localization, physical-size measurement error, and complete-pipeline speed, as summarized below.

Comparison of localization, physical-size measurement, and complete-pipeline speed

Method	Output	mAP@0.5 / %	mAP@0.5:0.95 / %	Overall MAE / mm	FPS
YOLOv5s	Rectangle	98.50	96.20	16.36	59.36 ± 0.72
YOLOv8s	Rectangle	98.80	96.50	15.73	55.61 ± 0.31
YOLOv10s	Rectangle	98.95	96.80	18.17	54.52 ± 0.21
YOLO11s	Rectangle	99.05	97.05	16.45	53.86 ± 0.57
YOLOv13-S	Rectangle	99.15	97.25	15.96	42.73 ± 0.81
RT-DETR-L	Rectangle	99.10	97.40	16.91	19.21 ± 0.14
YOLOv8s-seg + `fitEllipse`	Segmentation	99.00	97.10	21.79	51.79 ± 0.47
Proposed method	Ellipse	97.30	94.60	**3.17**	35.35 ± 0.88

For object localization, the recent rectangular detectors achieved mAP@0.5 values of 98.80–99.15% and mAP@0.5:0.95 values of 96.50–97.40%. These values were higher than the 97.30% and 94.60%, respectively, obtained by the proposed ellipse detector before GCER. Thus, in terms of conventional object-localization metrics, the recent rectangular detectors performed better than the proposed ellipse detector.

In physical-size measurement, the methods showed a clear difference. The recent box-based detectors produced overall size MAEs of 15.73–18.17 mm, with YOLOv8s giving the lowest error of 15.73 mm. YOLOv8s-seg followed by `fitEllipse` produced an overall MAE of 21.79 mm. By comparison, the complete proposed single-frame method achieved major-axis, minor-axis, and overall MAEs of 3.09, 3.25, and 3.17 mm, respectively, with an RMSE of 3.18 mm and a MAPE of 0.84%. Relative to YOLOv8s, the strongest box-based measurement baseline, the proposed method reduced the overall MAE by 79.85%.

This difference arises because the width and height of a conventional rectangular box describe the horizontal enclosing region, which can include background and do not directly correspond to the major and minor axes of the roller tank lug working contour. The box also changes with the visible region when the target is partially occluded or its boundary is incomplete. The proposed method directly predicts the ellipse center, semi-axes, and orientation, allowing its output to be used directly for size reconstruction. GWD jointly constrains the ellipse center, axis lengths, and orientation. During training, $\mathcal{L}_{\mathrm{ma}}$ converts the predicted ellipse dimensions into millimeters and penalizes their deviations from the manually measured physical dimensions. GCER then uses the remaining visible boundary evidence to correct residual offsets between the initial ellipse and the physical contour. The principal advantage of the proposed method therefore lies in physical-size measurement rather than in conventional box-localization metrics.

For complete-pipeline speed, all methods were tested on an NVIDIA RTX A6000 with a 640×640 input, a batch size of 1, and five repeated runs. The complete proposed single-frame pipeline achieved 35.35 ± 0.88 FPS. Its speed was lower than those of the lightweight YOLO detectors because the timing also included ellipse decoding, normal-direction edge search, rejection of inconsistent

candidates, ellipse refitting, and physical-size reconstruction. Nevertheless, the complete method remained above 35 FPS and met the real-time processing requirement of the mine-site server.

The engineering maturity of YOLOv5 was an additional consideration. Its stable open-source training and inference pipeline facilitates controlled changes to the output dimensionality, target construction, and loss formulation. Moreover, the original YOLOv5s contains 7.013 M parameters and requires 15.8 GFLOPs, while the modified ellipse detector contains 7.025 M parameters and requires 16.0 GFLOPs. The ellipse-aware modification therefore increases the parameter count by approximately 0.17% and the computational cost by approximately 1.27%. The current field implementation uses centralized processing on the mine server rather than camera-side embedded inference. Nevertheless, the limited model growth provides a practical basis for subsequent migration to Jetson-class devices or other industrial edge-computing platforms. Future work will evaluate model compression, mixed-precision inference, and hardware acceleration, and will assess edge deployment in other mines and related industrial measurement applications involving circular or elliptical components.

Finally, the proposed components are not structurally tied to YOLOv5. The ellipse head replaces the detector output interface, GWD and the measurement-aware objective operate at the loss level, and GCER performs local geometric refinement after detection. These components can therefore, in principle, be integrated into other YOLO variants or single-stage detectors. YOLOv5 was used to establish the method under a mature and computationally economical baseline, rather than being treated as the novelty or the only applicable backbone.

Response to Reviewer 3

R3.General There are several problems of English in the paper. Some examples are given below.

Response We sincerely thank the reviewer for drawing attention to the English presentation. We agree that the original manuscript contained unclear constructions, inconsistent abbreviation handling, and several long sentences that obscured the intended technical meaning. In response, we conducted a manuscript-wide language and terminology audit rather than correcting only the examples listed in the report. We revised sentence structure, grammar, hyphenation, and technical phrasing; standardized the terminology used for the detector, measurement-aware objective, geometric refinement, and wear-state assessment; and checked that abbreviations are defined at first use. These edits were made without changing the reported data or the technical conclusions. The specific corrections corresponding to the reviewer's examples are detailed below.

English-language comments

R3.Q1 Abstract: "Reliable online measurement is difficult because background clutter, partial occlusion, perspective projection, and non-frontal imaging can distort the visible lug boundary." In this sentence I think an "of" has been missed.

Response We thank the reviewer for carefully pointing out this language issue. We agree that the original Abstract did not explicitly identify the object of the online measurement, making the sentence read as incomplete. Following the reviewer's suggestion, we rechecked and rewrote the sentence to specify that the measured quantity is the roller tank lug dimensions, so that readers do not have to infer it from the surrounding context. The revised sentence is:

Reliable online measurement of roller tank lug dimensions remains difficult because background clutter, partial occlusion, and perspective projection can distort the visible lug boundary.

R3.Q2 Introduction: "These errors may be acceptable for category-level detection, but they directly propagate into millimeter level size estimation and wear classification." This statement seems to be incomplete. Not clear what author is trying to say here.

Response We sincerely thank the reviewer for pointing out this unclear statement. We agree that the original sentence stated only that localization errors affect millimeter-level size estimation and wear-state assessment, without explaining how the errors propagate from the image domain to the reconstructed physical dimensions. The phrase "wear classification" could also lead readers to assume that a separately trained wear classifier was used. Following the reviewer's comment, we reorganized this part of the Introduction to make the causal relationship explicit.

In the original sentence, "these errors" referred to geometric errors introduced by the rectangular representation, including the additional background enclosed by a rectangular box, displacement of the detected boundary with the visible region under partial occlusion, and inconsistency between

the box width and height and the actual working contour of the roller tank lug. In category-level detection, a prediction may still be regarded as correct when the box sufficiently overlaps the target and the class is correct, even if a small boundary displacement remains. In the present measurement task, however, the predicted geometric parameters are passed directly to the camera model and converted into the major and minor physical dimensions in millimeters. Errors in the semi-axes therefore propagate to the reconstructed dimensions, while errors in the ellipse center can also affect the off-axis correction and the local edge search performed by GCER. These errors consequently affect the final size measurement. Because the wear state is assigned using the 4-mm and 8-mm thresholds, even a small size error may change the final state for a sample close to either threshold.

We therefore rewrote the corresponding sentences in the Introduction and consistently replaced the potentially ambiguous term "wear classification" with "threshold-based wear-state assessment." This wording makes clear that the final state is assigned from the reconstructed dimensions using predefined thresholds rather than predicted by a second classification network.

Because the lug appears circular or elliptical in the image, a rectangular box necessarily includes additional background. Under partial occlusion or off-axis imaging, the box may also change with the visible region. A lug can therefore be detected correctly while its box-derived dimensions remain inaccurate, which may subsequently alter the assigned wear state. Roller tank lug monitoring thus requires a target representation that directly describes the working contour and supports physical size reconstruction.

Because the state assignment is threshold-based, a size-reconstruction error can change the assigned state when the estimated wear depth lies close to δ or ε. This sensitivity motivates improving the ellipse geometry even when the target has already been correctly detected at the category level.

R3.Q3 There are usage of abbreviations which has not been expanded or elaborated upon (GWD, MAE, GCER). There may be few more if we look closely.

Response We thank the reviewer for this reminder. To make the manuscript easier to read, we checked all abbreviations throughout the paper. Gaussian Wasserstein distance (GWD), mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), geometry-constrained ellipse refinement (GCER), and median absolute deviation (MAD), among others, are now written in full at their first occurrence, after which only the corresponding abbreviations are used consistently. We also checked the terminology used in figure captions, table captions, and algorithms. In addition, to avoid reusing the same symbol for different quantities, q_i is retained exclusively for frame-level detection confidence, whereas a point-level GCER candidate is denoted by $p_{k,r}$.

Gaussian Wasserstein distance (GWD) jointly constrains the ellipse geometry, while a differentiable analytical camera-metrology mapping converts the image-domain parameters into physical dimensions. A measurement-aware objective further constrains parameter errors according to their effects on millimeter-level size estimation. A geometry-constrained ellipse refinement (GCER) procedure

then uses local edge evidence to correct contour offsets, and robust multi-frame fusion suppresses unreliable frame-level measurements before threshold-based wear-state assessment.

Experiments on a dedicated roller tank lug dataset showed that the complete single-frame pipeline achieved an overall mean absolute error (MAE) of 3.17 mm.

The complete single-frame pipeline, including GCER, operated at 35.35 ± 0.88 frames per second (FPS).

We report major-axis mean absolute error (MAE), minor-axis MAE, overall MAE, root mean square error (RMSE), and mean absolute percentage error (MAPE).

R3.Q4 There are few more statements like this. I will leave it to the authors to correct in the final manuscript.

Response We thank the reviewer for this valuable suggestion. Rather than correcting only the examples identified in the review, we systematically revised the language throughout the manuscript. The revisions included splitting excessively long sentences, supplying missing subjects or logical connections, standardizing tense and article usage, removing repetitive statements, using technical terms such as "roller tank lug" consistently, and checking the grammatical transitions before and after equations.

Technical comments

R3.Q5 Not clear how contribution 3 is presented in this paper. If it is referring to multi-frame fusion, then authors should reword contribution 3.

Response We thank the reviewer for identifying that the third contribution was not presented with sufficient precision. We agree that the original contribution statement combined several operations too broadly and did not establish a clear correspondence between the claimed contribution and its methodological realization. This wording could make it unclear whether Contribution (3) referred to edge-guided ellipse refinement, physical-size reconstruction, or multi-frame fusion.

We have therefore reorganized the contribution list so that each item maps to a distinct method component and its corresponding experiment. Contribution (3) now refers exclusively to the geometry-constrained ellipse refinement procedure (GCER): boundary evidence is sampled around the initial ellipse, unreliable edge points are rejected, and a robust ellipse is refitted to reduce the discrepancy between the prediction and the visible lug contour. Contribution (4) separately describes physical-size reconstruction, robust multi-frame size fusion, and threshold-based wear-state assessment. The multi-frame module rejects abnormal frame-level measurements and adaptively weights retained measurements according to detection confidence, median consistency, and temporal stability. This restructuring removes the ambiguity noted by the reviewer and makes the contribution statements consistent with Sections II-D and III-B and their corresponding experiments.

(3) We develop GCER to search for valid edge evidence around the initial ellipse and refit the contour, improving boundary localization without introducing a second deep network.

(4) We integrate physical size reconstruction, robust multi-frame fusion, and threshold-based wear assessment into a single pipeline, reducing the influence of abnormal frames on the final dimensions and wear state.

R3.Q6 What is the need of class score. There is only one class to detect, isn't it?

Response We thank the reviewer for pointing out this unnecessary ambiguity. The reviewer is correct that the detector has only one semantic target class, namely the roller tank lug. Consequently, a separate class score does not perform multi-class discrimination in this application. The original notation listed objectness and class score separately because the modified head retained the standard YOLOv5 output interface, but this presentation was redundant for the single-class setting and could incorrectly suggest that the class score represented the normal, slight-wear, and severe-wear states. It does not.

To make the formulation consistent with the actual role of this output, we now combine the objectness and single target-class probability into one inference-time detection confidence, denoted by $\hat{c}_{\text{det}}$. The prediction vector has therefore been rewritten as $\hat{p} = (\hat{x}_c, \hat{y}_c, \hat{a}, \hat{b}, \hat{\theta}, \hat{c}_{\text{det}})$. This confidence indicates the reliability of the roller-tank-lug detection; it is used for candidate filtering and as the confidence component $q_i = \hat{c}_{\text{det},i}$ in robust multi-frame fusion. It is not a wear-state probability and does not determine the normal/slight/severe label.

We have also clarified the training objective. The classification term is retained in the general YOLOv5 loss notation for architectural completeness, but in the one-class implementation it does not perform inter-class discrimination. The final wear state is assigned only after physical-size reconstruction by applying the warning and replacement thresholds to the wear-related dimensional-deviation indicator. The detection-head notation, loss explanation, and fusion-confidence definition have all been revised accordingly.

For the present single-class setting, the standard objectness and target-class probability are combined at inference into one detection confidence. Each anchor prediction is therefore represented as

$$\hat{p} = (\hat{x}_c, \hat{y}_c, \hat{a}, \hat{b}, \hat{\theta}, \hat{c}_{\text{det}}),$$

where $\hat{c}_{\text{det}}$ is the confidence that the prediction is a roller tank lug. It is used for candidate filtering and frame-reliability weighting, and is not a normal/slight/severe wear probability.

$\mathcal{L}_{\text{cls}}$ follows the YOLOv5 loss definition. Because the detector has only one target class, this term does not distinguish wear states; the wear state is assigned later from the reconstructed dimensions.

where $\hat{c}_{\text{det},i}$ is the single-class detection confidence of frame i and is used only as a dimensionless frame-reliability score.

R3.Q7 Equation 9 and 10 needs clarification. What is ω exactly? Use of differential camera metrology is not clear. What feature or parameter of camera metrology is being referred to here?

Response We sincerely thank the reviewer for identifying that the definitions associated with the original Equations (9) and (10) were insufficient. We fully agree that the original text did not provide an explicit expression for ω and did not explain what was meant by "differentiable camera metrology." The wording could incorrectly suggest that ω was a learned parameter or that the metrology component extracted an additional camera feature. Rechecking the equations against the implementation also showed that the original sensitivity-loss notation was too broad: the implemented sensitivity term acts only on the two semi-axes and applies the sensitivity coefficient after computing the SmoothL1 error. We have corrected both the explanation and the formula.

The camera-metrology component is not an additional trainable neural-network layer. It is an analytical physical-size mapping implemented with differentiable tensor operations. For decoded semi-axes $\hat{a}$ and $\hat{b}$ in input-image pixels, the reconstructed dimensions are

$$\hat{W} = \frac{2d\hat{a}}{f_x}, \qquad \hat{H} = \frac{2d\hat{b}}{f_y}.$$

Here f_x and f_y are pre-calibrated focal lengths and d is the recorded camera-to-target distance. They are treated as known constants and are not updated during training. "Differentiable" means that the physical-size error computed from $\hat{W}$ and $\hat{H}$ remains connected to $\hat{a}$ and $\hat{b}$ in the computational graph, so the error can be back-propagated to the ellipse prediction head. No camera feature is learned or optimized by this mapping.

The sensitivity coefficient $\boldsymbol{\omega}_i$ is the analytical change in reconstructed physical size caused by a one-pixel change in the predicted semi-axis:

$$\boldsymbol{\omega}_i = \left[\left| \frac{\partial \hat{W}_i}{\partial \hat{a}_i} \right|, \left| \frac{\partial \hat{H}_i}{\partial \hat{b}_i} \right| \right] = \left[\frac{2d_i}{f_x}, \frac{2d_i}{f_y} \right].$$

Thus, $\omega_{i,a}$ and $\omega_{i,b}$ quantify how many millimeters of physical-size variation are induced by a one-pixel error in a and b, respectively. When a semi-axis is still represented in the coordinate system of detection level l, the implementation includes its stride and uses $[2s_l d_i/f_x,\ 2s_l d_i/f_y]$. The sensitivity coefficients are computed from known quantities and detached from gradient computation; they are not learned parameters.

To match the actual implementation, we replaced the original expression $\text{SmoothL1}(\omega \odot (\hat{E} - E_{gt}))$ with

$$\mathcal{L}_{\text{sens}} = \frac{1}{2N_v} \sum_{i=1}^{N_v} \sum_{k \in \{a,b\}} \omega_{i,k} \rho(\hat{e}_{i,k} - e^{gt}_{i,k}),$$

where ρ is SmoothL1, N_v is the number of matched targets with valid physical-size annotations, and $\hat{e}_i = (\hat{a}_i, \hat{b}_i)$. Only the two semi-axes are included because the near-frontal size mapping directly depends on a and b. The center and orientation remain constrained by the coarse ellipse-regression

term and GWD. The complete measurement-aware objective remains $\mathcal{L}_{\text{ma}} = \mathcal{L}_{\text{size}} + \alpha\mathcal{L}_{\text{sens}}$, with $\alpha = 0.001$ in the reported experiment. We have revised Section II-C and the experimental settings in Section IV-A accordingly, and have replaced "camera metrology layer" with "differentiable analytical camera-metrology mapping" throughout the manuscript.

$\mathcal{L}_{\text{ma}}$ converts the predicted semi-axes into physical dimensions through the analytical camera model and penalizes the resulting size error. This mapping is implemented using differentiable tensor operations and contains no trainable parameters. The calibrated focal lengths and recorded distance are treated as known constants and are not updated during training.

$$\boldsymbol{\omega}_i = \left[\left|\frac{\partial \hat{W}_i}{\partial \hat{a}_i}\right|, \left|\frac{\partial \hat{H}_i}{\partial \hat{b}_i}\right|\right] = \left[\frac{2d_i}{f_x}, \frac{2d_i}{f_y}\right].$$

Thus, each component of $\boldsymbol{\omega}_i$ expresses the physical-size change, in millimeters, caused by a one-pixel error in the corresponding semi-axis. For a prediction expressed on detection level l, the implementation uses $\boldsymbol{\omega}_i^{(l)} = [2s_l d_i/f_x,\ 2s_l d_i/f_y]$ before pixel-coordinate decoding. The coefficients are computed from d_i, f_x, f_y, and the detection-level stride and are detached from gradient computation; they are neither learned by the network nor obtained from image features.

$$\mathcal{L}_{\text{sens}} = \frac{1}{2N_v}\sum_{i=1}^{N_v}\sum_{k\in\{a,b\}} \omega_{i,k}\rho(\hat{e}_{i,k} - e_{i,k}^{gt}), \qquad \mathcal{L}_{\text{ma}} = \mathcal{L}_{\text{size}} + \alpha\mathcal{L}_{\text{sens}}.$$

where $\hat{e}_i = (\hat{a}_i, \hat{b}_i)$. Only the semi-axis errors are included because the near-frontal size mapping directly depends on a and b. The ellipse center and orientation remain constrained by the coarse ellipse-regression term and GWD. The measurement-aware objective is therefore

$$\mathcal{L}_{\text{ma}} = \mathcal{L}_{\text{size}} + \alpha\mathcal{L}_{\text{sens}},$$

where α balances direct physical-size supervision and sensitivity-weighted semi-axis regression. This formulation follows the actual implementation and preserves the efficiency of YOLOv5 while making the training objective consistent with the final engineering requirement.

R3.Q8 I think equation 21 is wrong. Why is 2 at the subscript? Or there is some error which I did not get.

Response We thank the reviewer for identifying this unclear notation. The subscript 2 denotes the Euclidean (L_2) norm and is not a frame index. For the two-dimensional reconstructed size $S_i = [W_i, H_i]^\top$ and the component-wise median $S_m = [W_m, H_m]^\top$, the scalar deviation is

$$\Delta_i^{\text{med}} = \|S_i - S_m\|_2 = \sqrt{(W_i - W_m)^2 + (H_i - H_m)^2}.$$

This deviation has the same physical unit as W and H. The robust scale and retained-frame set are then defined as

$$\mathrm{MAD} = \mathrm{median}\left(\{\Delta_i^{\mathrm{med}}\}_{i=1}^{N}\right), \qquad \Omega = \left\{ i \,\middle|\, \Delta_i^{\mathrm{med}} \le \tau_m(\mathrm{MAD} + \epsilon) \right\}.$$

For each retained frame, the median-consistency weight is $r_i = \exp[-\Delta_i^{\mathrm{med}}/(\mathrm{MAD} + \epsilon)]$. Because the numerator and denominator have the same unit, their ratio and hence r_i are dimensionless. In the revised manuscript, Δ_i^{med} is described concisely as the Euclidean size deviation in millimeters; the explicit expansion is provided here to answer the notation question directly.

where Δ_i^{med} is the Euclidean distance between the frame-level size and the component-wise median. It measures the joint deviation of W and H and is expressed in millimeters.

R3.Q9 What is the equation for computation of the overall error? How it is being reported in mm?

Response We thank the reviewer for identifying that the original manuscript reported "overall MAE" without providing its explicit definition. The reported value is an end-to-end physical measurement error, calculated after the predicted ellipse parameters have been converted into the reconstructed major and minor dimensions, $\hat{W}_i$ and $\hat{H}_i$, using the calibrated camera-metrology model. These reconstructed dimensions are compared directly with the corresponding manually measured ground-truth dimensions, W_i^{gt} and H_i^{gt}. For N test samples, the overall MAE is

$$\mathrm{MAE}_{\mathrm{overall}} = \frac{1}{2N} \sum_{i=1}^{N} \left(\left| \hat{W}_i - W_i^{gt} \right| + \left| \hat{H}_i - H_i^{gt} \right| \right).$$

Equivalently, this is the arithmetic mean of the separately reported major-axis and minor-axis MAEs. For example, the complete single-frame method has major- and minor-axis MAEs of 3.09 and 3.25 mm, respectively, giving $(3.09 + 3.25)/2 = 3.17$ mm. The corresponding RMSE and MAPE are calculated over the same $2N$ reconstructed dimensions. The error is reported in millimeters because both the reconstructed dimensions and the manually measured ground-truth dimensions are physical lengths in millimeters; their absolute differences therefore retain the unit millimeter. MAPE is dimensionless and is reported as a percentage. We have added the explicit definitions of overall MAE, RMSE, and MAPE to the dimension-measurement subsection of the revised manuscript.

To determine whether accurate localization also yields accurate dimensions, we compared conventional image processing, recent detectors with box-derived dimensions, and ellipse-based methods under the same test split and camera calibration. YOLOv8s-seg followed by `fitEllipse` served as a segmentation-and-fitting baseline. We report major-axis mean absolute error (MAE), minor-axis MAE, overall MAE, root mean square error (RMSE), and mean absolute percentage error (MAPE).

For sample i, let $\hat{W}_i$ and $\hat{H}_i$ denote the reconstructed dimensions and W_i^{gt} and H_i^{gt} the manually measured references. The overall metrics are

$$\mathrm{MAE}_{\mathrm{overall}} = \frac{1}{2N}\sum_{i=1}^{N}\left(|\hat{W}_i - W_i^{gt}| + |\hat{H}_i - H_i^{gt}|\right),$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{2N}\sum_{i=1}^{N}\left[(\hat{W}_i - W_i^{gt})^2 + (\hat{H}_i - H_i^{gt})^2\right]},$$

$$\mathrm{MAPE} = \frac{100}{2N}\sum_{i=1}^{N}\left(\frac{|\hat{W}_i - W_i^{gt}|}{W_i^{gt}} + \frac{|\hat{H}_i - H_i^{gt}|}{H_i^{gt}}\right).$$

Accordingly, $\mathrm{MAE}_{\mathrm{overall}} = (\mathrm{MAE}_{\mathrm{major}} + \mathrm{MAE}_{\mathrm{minor}})/2$. Because both predictions and ground truth are expressed in millimeters after physical size reconstruction, MAE and RMSE are reported in millimeters, whereas MAPE is reported as a percentage.

R3.Q10 More details of the training, validation, and testing data has to be provided. Will the authors share this data?

Response We thank the reviewer for emphasizing dataset transparency and reproducibility. We agree that, although the original manuscript reported the number of samples in each subset, it did not sufficiently describe the acquisition conditions, the unit used for dataset partitioning, the annotation workflow, or the acquisition of the physical ground truth. We have added this information to the revised manuscript.

The dataset contains 7,200 images and 11,274 roller tank lug instances. The training, validation, and test sets contain 5,040, 1,440, and 720 images, respectively. Because a single image may contain more than one roller tank lug, the instance counts exceed the image counts. To avoid mixing image-level and instance-level statistics, Table I now reports the numbers of images and annotated instances for each subset separately. The samples cover roller tank lugs of different nominal specifications and physical identities, multiple acquisition periods, different hoisting units and installation positions, and images acquired by multiple cameras under different viewing distances, illumination conditions, dust interference, and degrees of occlusion. The lug contours were first annotated as polygons in CVAT and then converted into initial ellipses using `fitEllipse`. Every fitted ellipse was subsequently inspected and corrected manually to ensure agreement with the working contour. The physical major and minor dimensions and camera-to-target distances were repeatedly measured by different field personnel, and the averages of the valid measurements were used as the physical reference values.

The training, validation, and test sets were partitioned into mutually exclusive groups according to the physical roller tank lug identity, acquisition period, and source video or continuous acquisition sequence. Each group was assigned in its entirety to only one subset. This procedure prevents the same physical lug or adjacent frames from appearing in different subsets and avoids object-identity

and temporal information leakage.

Regarding data sharing, we fully understand the importance of public data for independent reproduction. The dataset is currently managed in cooperation with the participating mine. Public release of the images and associated records requires formal authorization from the mine's electromechanical department and the relevant management units, together with screening and de-identification of sensitive site information. We therefore cannot release the complete dataset without the mine's permission. We are currently seeking the required authorization and preparing the material that can be released.

Upon acceptance, we will promptly release the implementation code, model configurations, annotation-conversion utilities, size-measurement and multi-frame fusion programs, and usage documentation. For the dataset, we will release all images, annotations, camera-calibration information, and physical measurement records for which authorization has been obtained and de-identification has been completed. Because the manuscript is currently under double-anonymous review, the existing repository metadata and industrial authorization documents may disclose the identities of the authors or the participating site. We therefore do not provide a public repository containing such identifying information during the present review stage.

R3.Q11 Most fundamental problem I have with this paper is the following: Why a two stage detector? First an ellipse detector, size estimator, and then wear state classifier for the lug. A well integrated single deep learning architecture could have do the same work faster and better with appropriate training data.

Response We thank the reviewer for raising this fundamental architectural question. We agree that, from the perspective of the processing flow, the proposed method can be understood as two stages: the first stage uses the ellipse-aware detector to obtain the roller tank lug geometry and reconstruct its physical dimensions, and the second stage determines the wear state from the fused dimensions. The original manuscript did not adequately explain why this staged design was adopted, which could make a direct end-to-end three-class network appear more straightforward. We have revised the explanation accordingly.

We adopted this two-stage flow because the required output is not limited to the three discrete labels of normal, slight wear, and severe wear. Field maintenance also requires continuous millimeter-level results, including the major dimension, minor dimension, and dimensional deviation. These quantities allow operators to verify the basis of a state decision, inspect samples close to a maintenance threshold, and track dimensional changes over time. The first stage provides these measurable physical quantities, after which the second stage assigns the state using the 4-mm warning threshold and the 8-mm replacement threshold. If the thresholds change for a different roller tank lug specification or maintenance policy, only the decision rules need to be updated; the complete image dataset does not need to be relabeled to train a new three-class model. The resulting state decision can also be traced back to the measured physical dimensions rather than being represented only

by a class probability.

In implementation, the two-stage flow does not mean that two large deep networks are executed sequentially. The ellipse-aware YOLOv5 detector is the only learned inference module. GCER, camera-based size conversion, multi-frame fusion, and threshold-based wear-state assessment are deterministic geometric, statistical, and rule-based operations. On the RTX A6000 server used at the mine site, the complete single-frame pipeline achieved 35.35 ± 0.88 FPS, while fusion over a 30-frame window added only 0.16 ms. The staged processing therefore did not prevent the system from operating in real time.

We also agree that the end-to-end integrated architecture suggested by the reviewer is a valuable research direction. A unified multi-task network combining ellipse regression, continuous size estimation, and wear-state prediction may reduce intermediate processing and enable joint optimization of the complete task. We will investigate such an end-to-end multi-task architecture in future work.

(4) We integrate physical size reconstruction, robust multi-frame fusion, and threshold-based wear assessment into a single pipeline, reducing the influence of abnormal frames on the final dimensions and wear state.

As shown in Fig. 2, the framework contains four stages. First, an ellipse-aware YOLOv5 detector outputs an initial ellipse E_0 and an object confidence score. Second, a measurement-aware objective guides training by penalizing physical size errors after camera projection. Third, an edge-guided refinement step adjusts E_0 using local boundary evidence and robust ellipse refitting. Finally, physical dimensions are reconstructed and fused across adjacent frames before threshold-based wear-state assignment. During inference, the ellipse detector is the only learned module; contour refinement, size reconstruction, multi-frame fusion, and wear-state assignment are performed by deterministic geometric, statistical, and threshold operations.

This staged design retains the major and minor physical dimensions and the dimensional deviation, rather than producing only a discrete wear label. These millimeter-level quantities allow the decision to be checked and dimensional changes to be tracked over time. They also separate the maintenance rule from network training: when the nominal lug specification or maintenance thresholds change, the decision rule can be updated without relabeling the image dataset and retraining a three-class model.

The wear depth is computed directly from the fused physical size and the nominal lug diameter D:

$$\text{Wear} = \max\left(|D - \bar{W}|, |D - \bar{H}|\right).$$

The wear state is assigned by two thresholds:

$$State = \begin{cases} \text{Normal}, & \text{Wear} < \delta, \\ \text{Slight}, & \delta \leq \text{Wear} < \varepsilon, \\ \text{Severe}, & \text{Wear} \geq \varepsilon. \end{cases}$$

R3.Q12 Authors may consider the following paper on ellipse detection and cite it: "Pre-processing by data augmentation for improved ellipse fitting," Pankaj Kumar, Erika R. Belchamber, and Stanley J. Miklavcic, `https://doi.org/10.1371/journal.pone.0196902`.

Response We thank the reviewer for recommending this closely related ellipse-fitting study. We carefully read and cited the work of Kumar et al. Their method augments the input point set according to point eccentricity and improves the accuracy of conventional ellipse fitting, including when the available point set is incomplete. This study provides useful background for the proposed GCER. Kumar et al. preprocess an existing point set, whereas GCER starts from a learned initial ellipse, searches for image evidence along local normal directions, rejects abnormal candidates, and refits the ellipse. We have added the citation and this relationship at the beginning of the Edge-Guided Ellipse Refinement subsection.

Ellipse-fitting accuracy can be improved by preprocessing the input point set. Kumar et al. replicated points in high-eccentricity regions before fitting and reported reduced errors for several conventional ellipse estimators. This work provides relevant context for the proposed geometry-constrained ellipse refinement (GCER). While their method augments an existing point set, GCER starts from a learned initial ellipse, searches for image evidence along local normal directions, rejects candidates with abnormal radial offsets, and refits the ellipse using the retained points.

Response to Reviewer 4

R4.Q1 The dataset is collected from a single industrial environment. The authors should discuss whether the model generalizes to different mines, lighting conditions, cameras, and roller tank lug designs.

Response We thank the reviewer for raising this important question about cross-mine generalization. The field data used for evaluation were collected at one mine, but not from a single hoisting conveyance or under a fixed imaging condition. Multiple cameras installed at different positions and heights monitored roller tank lugs on different hoisting conveyances. The dataset therefore covers different physical lugs, nominal specifications, camera configurations, acquisition positions, camera-to-target distances, and illumination conditions within the current mine.

Each camera, lens, and resolution configuration was calibrated separately using Zhang's method before acquisition. The focal length and image resolution of each camera remained fixed during its corresponding acquisition sessions, and each sample was reconstructed using the intrinsics of its source camera and the recorded camera-to-target distance. A change in the camera, lens, resolution, or imaging configuration therefore requires recalibration, but not a change to the reconstruction equations.

The present dataset already includes images acquired under different illumination conditions. For extremely low illumination, our group has previously developed a dedicated low-light enhancement method for roller tank lug images. This method can be used as an upstream image-enhancement step before the detection and measurement pipeline to improve the visibility of the lug contour.

Roller tank lugs are standardized guiding components in vertical-shaft hoisting systems. Although their nominal dimensions and maintenance thresholds may differ, their working contours share an annular geometry that can be represented as ellipses under camera projection. The ellipse prediction head estimates geometric parameters rather than matching a fixed appearance template. For a different roller tank lug specification, the nominal dimensions and maintenance thresholds can be updated without changing the ellipse detection, boundary refinement, or multi-frame fusion procedure. The ellipse prediction, GWD, measurement-aware supervision, and GCER components respectively constrain global ellipse geometry, physical-size error, and local boundary location; none of these formulations is tied to one particular camera, nominal specification, or illumination setting. These properties provide a technical basis for transferring the method to other mines.

We nevertheless fully agree that within-mine variation cannot replace independent cross-mine validation. The present study has not evaluated the framework using data from another mine, and we therefore do not claim that cross-mine generalization has been experimentally demonstrated. Following the reviewer's suggestion, this limitation is now stated explicitly in the revised Conclusion. Future work will collect independent data from additional mines and evaluate the method under different equipment configurations and operating environments.

The present study has not yet evaluated the framework using independent data from another mine,

and cross-mine generalization has therefore not been experimentally established.

R4.Q2 The study compares mainly against YOLOv5 variants. Comparisons with more recent detectors (e.g., YOLOv8/YOLOv10/RT-DETR) would better demonstrate the competitiveness of the proposed framework.

Response We thank the reviewer for this important and constructive suggestion. We fully agree that the original manuscript compared the proposed method mainly with the YOLOv5 rectangular baseline and its ellipse-aware variants. Although these controlled comparisons isolate the effects of the ellipse representation, GWD, $\mathcal{L}_{\mathrm{ma}}$, and GCER, they are insufficient to establish competitiveness against recent general-purpose detectors. We therefore added YOLOv8s, YOLOv10s, YOLO11s, YOLOv13-S, and RT-DETR-L. We also included YOLOv8s-seg followed by `fitEllipse` as a baseline that first segments the contour and then fits an ellipse. All methods were evaluated in terms of object localization, physical-size error, and complete-pipeline speed.

For object localization, the recent rectangular detectors achieved mAP@0.5 values of 98.80–99.15% and mAP@0.5:0.95 values of 96.50–97.40%. These values were higher than the 97.30% and 94.60%, respectively, obtained by the proposed ellipse detector before GCER. Thus, in terms of conventional object-localization metrics, the recent rectangular detectors performed better than the proposed ellipse detector.

The results changed substantially when the output was evaluated as a physical measurement. The recent box-based detectors produced overall size MAEs of 15.73–18.17 mm, with YOLOv8s giving the lowest error of 15.73 mm. YOLOv8s-seg followed by `fitEllipse` produced an overall MAE of 21.79 mm. By comparison, the complete proposed single-frame method achieved major-axis, minor-axis, and overall MAEs of 3.09, 3.25, and 3.17 mm, respectively, with an RMSE of 3.18 mm and a MAPE of 0.84%. Relative to YOLOv8s, the strongest box-based measurement baseline, the proposed method reduced the overall MAE by 79.85%.

This difference arises because the width and height of a conventional rectangular box describe the horizontal enclosing region, which can include background and do not directly correspond to the major and minor axes of the roller tank lug working contour. The box also changes with the visible region when the target is partially occluded or its boundary is incomplete. The proposed method directly predicts the ellipse center, semi-axes, and orientation, allowing its output to be used directly for size reconstruction. GWD jointly constrains the ellipse center, axis lengths, and orientation, thereby reducing geometric inconsistency among separately regressed parameters. During training, $\mathcal{L}_{\mathrm{ma}}$ converts the predicted ellipse dimensions into millimeters and penalizes their deviations from the manually measured physical dimensions. The model is therefore trained not only to locate the contour in pixels but also to reduce the final size error. GCER then uses the remaining visible boundary evidence to correct residual offsets between the initial ellipse and the physical contour. The principal advantage of the proposed method consequently lies in physical-size measurement rather than in conventional box-localization metrics.

For complete-pipeline speed, all methods were tested on an NVIDIA RTX A6000 with a 640×640 input, a batch size of 1, and five repeated runs. YOLOv5s, YOLOv8s, YOLOv10s, YOLO11s, and YOLOv13-S achieved 59.36 ± 0.72, 55.61 ± 0.31, 54.52 ± 0.21, 53.86 ± 0.57, and 42.73 ± 0.81 FPS, respectively. YOLOv8s-seg + `fitEllipse` achieved 51.79 ± 0.47 FPS, whereas RT-DETR-L achieved 19.21 ± 0.14 FPS. The complete proposed single-frame pipeline achieved 35.35 ± 0.88 FPS. Its speed was lower than those of the lightweight YOLO detectors because its timing included ellipse decoding, normal-direction edge search, rejection of inconsistent candidates, ellipse refitting, and physical-size reconstruction in addition to neural-network inference. These geometric operations add runtime without substantially increasing the network parameter count or GFLOPs. Nevertheless, the proposed method remained above 35 FPS and therefore met the real-time processing requirement of the mine-site server. Multi-frame fusion added only 0.16 ms per 30-frame window, and the complete pipeline required an average server-side computation time of 28.30 ms per frame.

Comparison of localization, physical-size measurement, and complete-pipeline speed

Method	Output	mAP@0.5 / %	mAP@0.5:0.95 / %	Overall MAE / mm	FPS
YOLOv5s	Rectangle	98.50	96.20	16.36	59.36 ± 0.72
YOLOv8s	Rectangle	98.80	96.50	15.73	55.61 ± 0.31
YOLOv10s	Rectangle	98.95	96.80	18.17	54.52 ± 0.21
YOLO11s	Rectangle	99.05	97.05	16.45	53.86 ± 0.57
YOLOv13-S	Rectangle	99.15	97.25	15.96	42.73 ± 0.81
RT-DETR-L	Rectangle	99.10	97.40	16.91	19.21 ± 0.14
YOLOv8s-seg + `fitEllipse`	Segmentation	99.00	97.10	21.79	51.79 ± 0.47
Proposed method	Ellipse	97.30	94.60	**3.17**	35.35 ± 0.88

For the proposed method, mAP was calculated from the ellipse detector before GCER, whereas the overall MAE and FPS correspond to the complete single-frame pipeline including GCER and physical-size reconstruction. The runtime of each comparison method likewise includes its required output processing.

For the proposed method, mAP was calculated from the ellipse detector before GCER, whereas the overall MAE and FPS correspond to the complete single-frame pipeline including GCER and physical-size reconstruction.

Overall, recent rectangular detectors retained an advantage in conventional localization metrics. The proposed method, however, substantially reduced the physical-size measurement error while maintaining real-time processing, allowing the detections to support more accurate millimeter-level measurement and wear-state assessment and thereby provide more reliable evidence for the safe operation of the hoisting system.

Physical-size reconstruction requires reliable localization. We therefore compared the proposed ellipse detector with recent one-stage and transformer-based detectors on the same test split. P, R, F1, mAP@0.5, and mAP@0.5:0.95 evaluate localization; parameter count and GFLOPs describe network complexity; and peak GPU memory and server-side end-to-end FPS include method-specific post-processing. FPS was averaged over five runs with a batch size of 1 and a 640×640

input.

Table II shows that the recent rectangular detectors achieved similar localization performance, with mAP@0.5 values of 98.80–99.15% and mAP@0.5:0.95 values of 96.50–97.40%. The ellipse detector obtained 97.30% and 94.60%, respectively. Although slightly lower under box-overlap metrics, it directly supplies the center, semi-axes, and orientation required for size reconstruction rather than recovering them indirectly from a rectangle.

The complete method achieved 35.35 ± 0.88 server-side end-to-end FPS, or 28.29 ms per frame. MAD rejection and adaptive weighted fusion added only 0.16 ms per 30-frame window, giving an amortized server-side cost of 28.30 ms per frame and approximately 28.45 ms from arrival of the final frame to the fused output.

YOLOv8s-seg followed by `fitEllipse` yielded 21.79 mm. The segmentation objective optimizes mask overlap rather than millimeter-level boundary position; jagged edges, gaps, and local shifts caused by dust, occlusion, or weak contrast therefore propagate into ellipse fitting. Box-based dimensions are similarly determined by the visible extrema and change when part of the contour is missing.

GWD then constrained the coupled ellipse parameters and reduced the MAE to 5.34 mm. $\mathcal{L}_{\mathrm{ma}}$ placed greater emphasis on image-domain errors with larger physical-size consequences, lowering the MAE to 4.33 mm. Finally, GCER used the visible boundary to correct residual local offsets and reduced the MAE to 3.17 mm.

R4.Q3 Although FPS is reported, additional information regarding GPU memory usage, parameter count, and computational complexity (GFLOPs) would help evaluate deployment feasibility.

Response We thank the reviewer for this practically important suggestion. We fully agree that FPS alone is insufficient to assess deployment feasibility, because model size, computational cost, runtime memory, and complete-pipeline latency jointly determine whether a method can operate on an industrial computing platform. We therefore added parameter count, GFLOPs, peak GPU memory, and server-side end-to-end FPS averaged over five repeated runs under a unified test protocol. All methods were evaluated directly on the mine-site server, which was equipped with an NVIDIA RTX A6000 GPU and an Intel Xeon Silver 4210R CPU, using a 640×640 input and a batch size of 1.

YOLOv5s contained 7.013 M parameters, required 15.8 GFLOPs, and occupied 0.76 GiB of peak GPU memory. The proposed ellipse detector contained 7.025 M parameters, required 16.0 GFLOPs, and occupied 0.77 GiB. Relative to YOLOv5s, the parameter count increased by only approximately 0.17%, GFLOPs by approximately 1.27%, and peak memory by 0.01 GiB. These results show that the ellipse prediction head does not substantially increase the neural network's resource requirements.

Computational complexity and runtime comparison

Method	Parameters / M	GFLOPs	Peak GPU memory / GiB	FPS
YOLOv5s	7.013	15.8	0.76	59.36 ± 0.72
YOLOv8s	11.136	28.6	0.81	55.61 ± 0.31
YOLOv10s	8.067	24.8	0.78	54.52 ± 0.21
YOLO11s	9.428	21.5	0.83	53.86 ± 0.57
YOLOv13-S	9.553	21.6	0.90	42.73 ± 0.81
RT-DETR-L	32.808	108.0	1.02	19.21 ± 0.14
YOLOv8s-seg + `fitEllipse`	11.790	40.2	0.84	51.79 ± 0.47
Proposed full method	7.025	16.0	0.77	35.35 ± 0.88

The parameter count and GFLOPs reported for the proposed method describe its learned ellipse detector, whereas its peak memory and FPS were measured for the complete single-frame pipeline. The complete method achieved 35.35 ± 0.88 FPS, compared with 59.36 ± 0.72 FPS for YOLOv5s. This reduction is mainly because the complete timing additionally includes ellipse decoding, GCER normal-direction edge search, robust candidate rejection, ellipse refitting, and physical-size reconstruction. The 30-frame fusion stage required only 0.16 ms per window. The amortized server-side computation time was 28.30 ms per frame, and the time from arrival of the final frame to the fused output was approximately 28.45 ms. Therefore, on the A6000 centralized server used in this study, the peak memory of 0.77 GiB and processing speed above 35 FPS demonstrate that the complete pipeline can operate in real time within the available hardware resources.

Training, testing, and field inference were performed directly on the mine-site workstation, equipped with an Intel Xeon Silver 4210R CPU at 2.40 GHz and an NVIDIA RTX A6000 GPU under Windows 10. Cameras at the surface and underground monitoring positions transmit images to this workstation through Ethernet and optical-fiber links for centralized processing.

Runtime was measured with a 640×640 input and a batch size of 1 over five repeated runs, including the post-processing required by each method.

The complete method achieved 35.35 ± 0.88 server-side end-to-end FPS, or 28.29 ms per frame. MAD rejection and adaptive weighted fusion added only 0.16 ms per 30-frame window, giving an amortized server-side cost of 28.30 ms per frame and approximately 28.45 ms from arrival of the final frame to the fused output.

R4.General The manuscript presents a practical and technically interesting contribution to industrial visual inspection. The measurement-aware optimization framework is novel and well aligned with engineering requirements. The experimental results demonstrate consistent improvements in measurement accuracy and wear-state classification. After addressing the above comments—particularly regarding generalization, reproducibility, and comparison with more recent detection architectures—the manuscript would be suitable for publication in IEEE Transactions on Industrial Informatics.

Response We sincerely thank the reviewer for the positive assessment of the engineering value

and measurement-aware design of this work. We made revisions in the three areas emphasized in the general comment. First, we distinguished variation within the current mine from cross-mine generalization, which has not yet been independently verified, and clarified the need for further evaluation across additional mines and their camera and roller tank lug configurations. Second, we expanded the descriptions of data acquisition, mutually exclusive group-wise dataset partitioning, annotation quality control, physical ground-truth measurement, training settings, and module parameters to improve experimental reproducibility. Third, we added recent YOLO models, RT-DETR-L, and a segmentation-followed-by-ellipse-fitting measurement baseline, and reported both conventional localization metrics and physical-size measurement errors under a unified protocol. These revisions strengthen the experimental evidence and define the scope of the method more clearly.